Revealing the physics of reionization with high-redshift quasars, JWST observations, & simulations of the IGM

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DUSC Seminar at the University of Washington
April 29, 2025

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What is reionization & why is it important?

- Ionization of the IGM by the first ionizing photon sources
- Took place at $z \sim 15 - 6(?)$
- Midpoint of $z \sim 7.5(?)$ (Planck)
- Driven by the first stars? Galaxies? AGN?

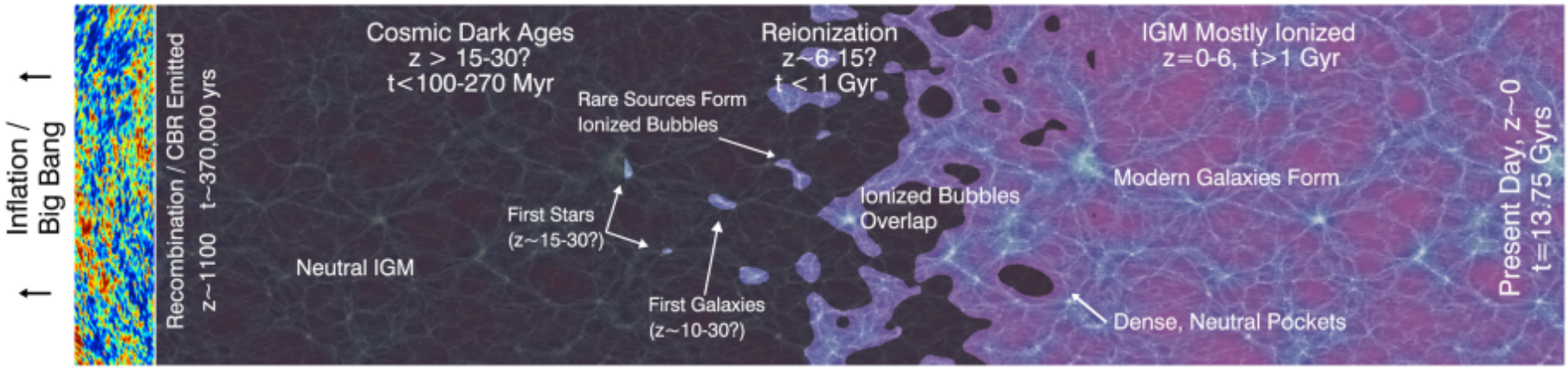


Figure: Robertson+10

Evidence for reionization

- Transmission in the Lyα Forest (Fan+06, Becker+15, ...)
- Transmission requires local $x_{\rm HI} \sim 10^{-5}$ at z = 6

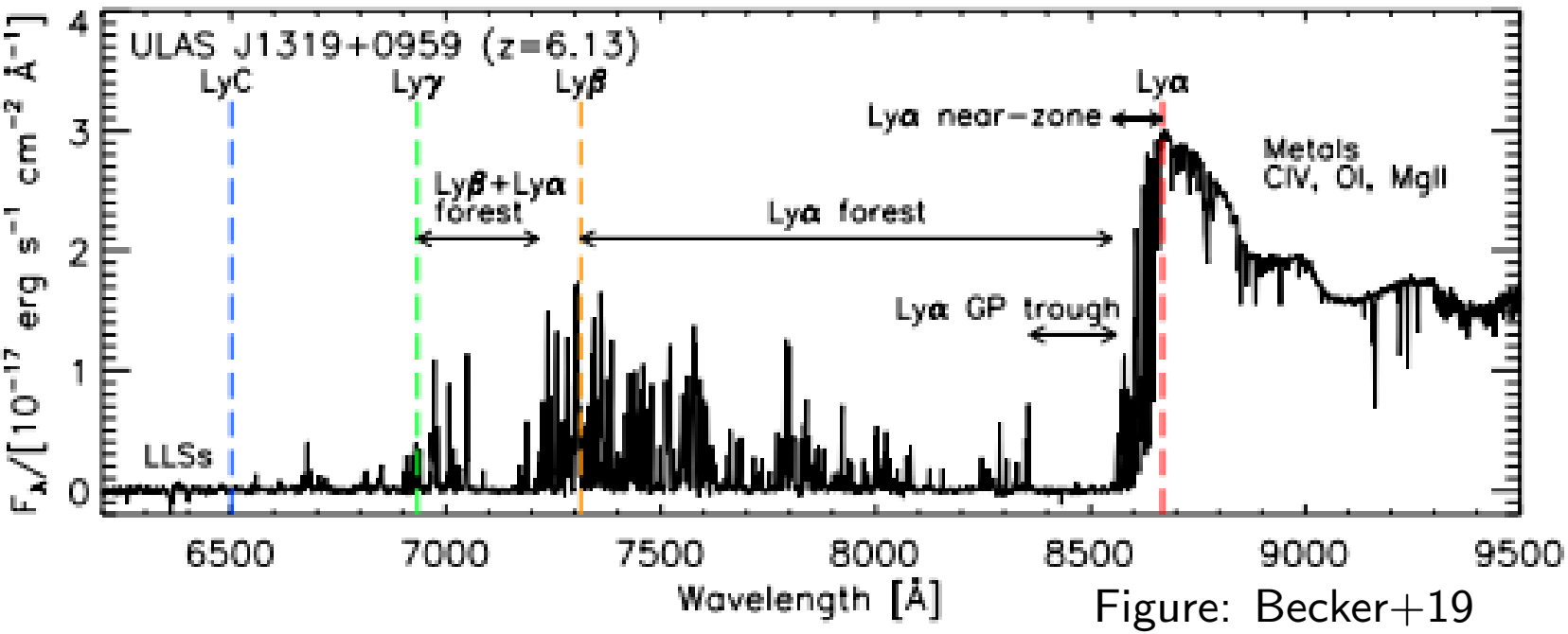


Figure: Becker+19

A critical juncture in IGM evolution

- IGM conditions changed radically during reionization
- Example: IGM temperature evolution

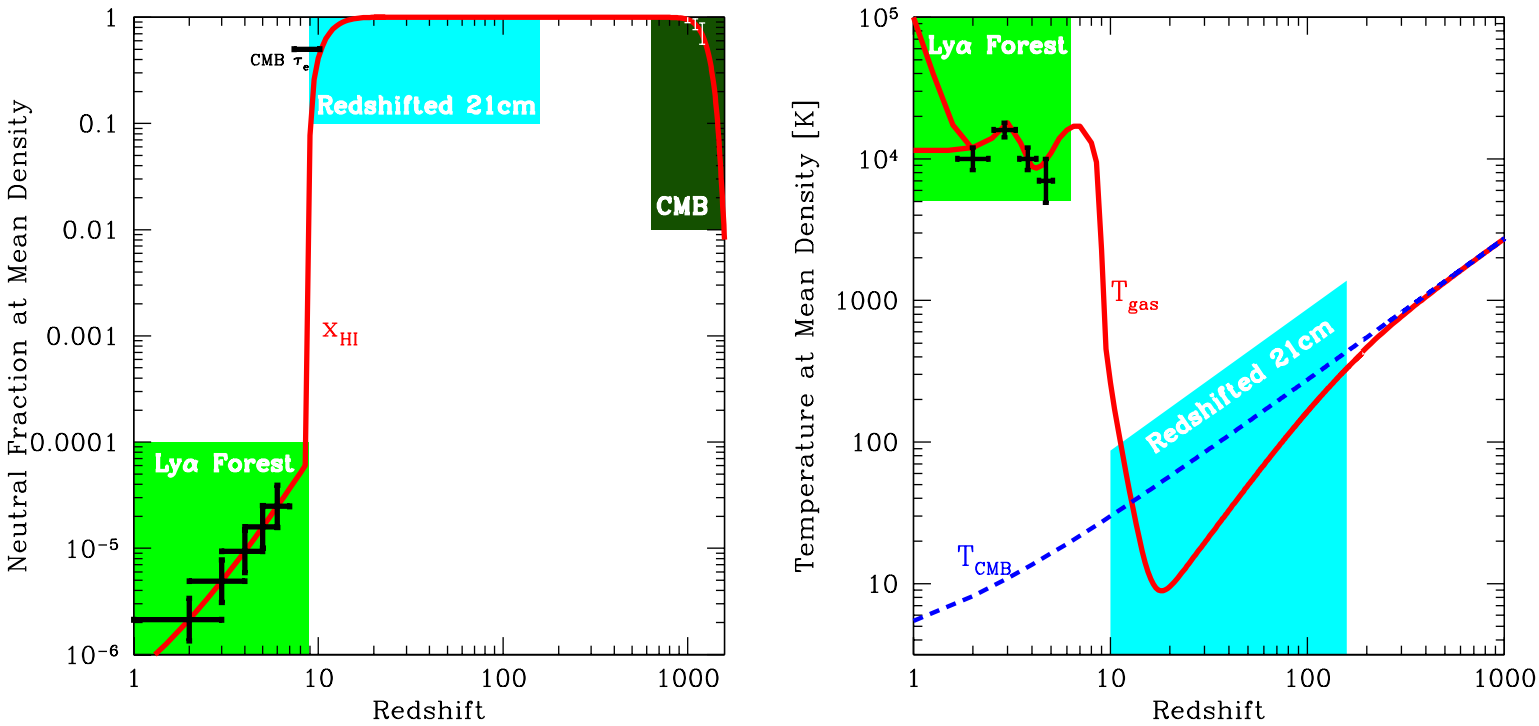
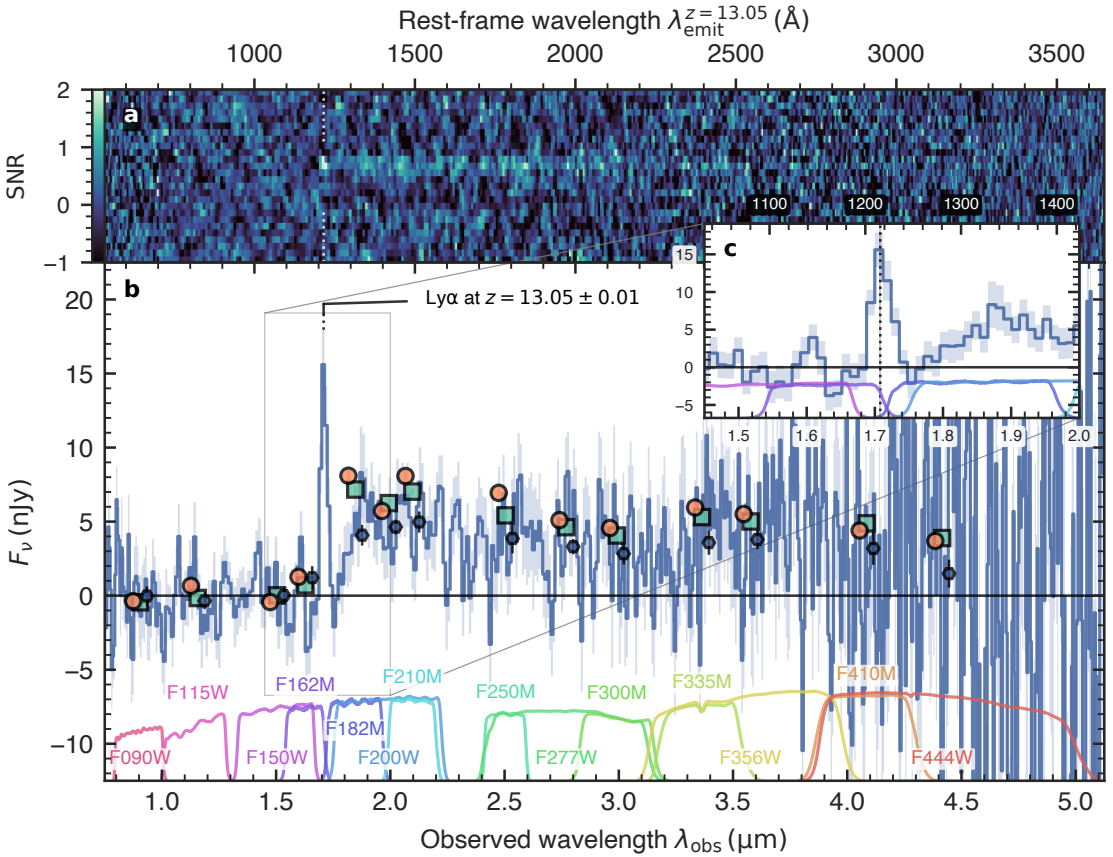
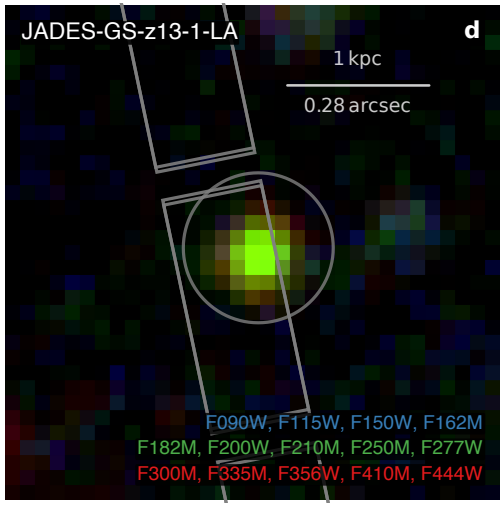


Figure: McQuinn 2016

Necessary context for high-redshift observations

- *Many* high-redshift observables are sensitive to reionization
- Example: $z = 13$ Lyα emitter?? (Witstok+24)

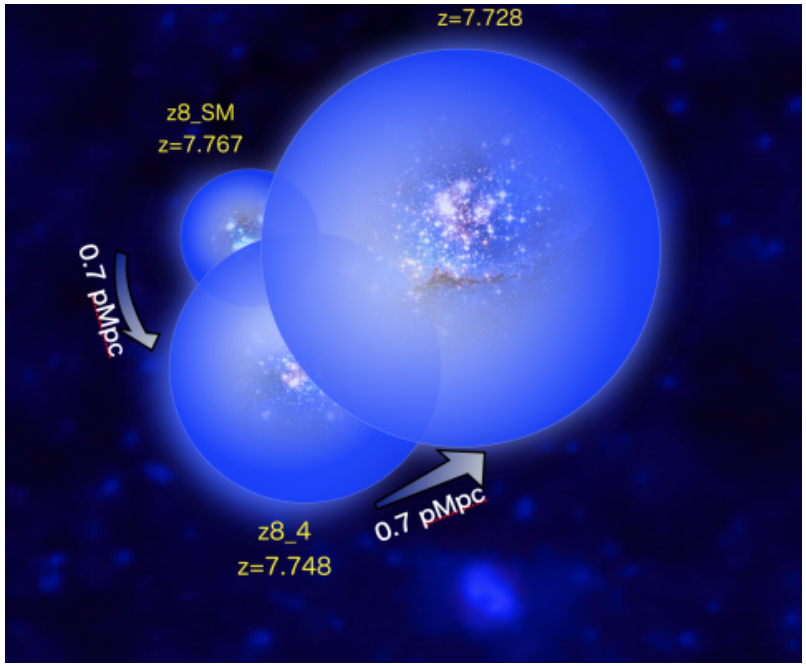


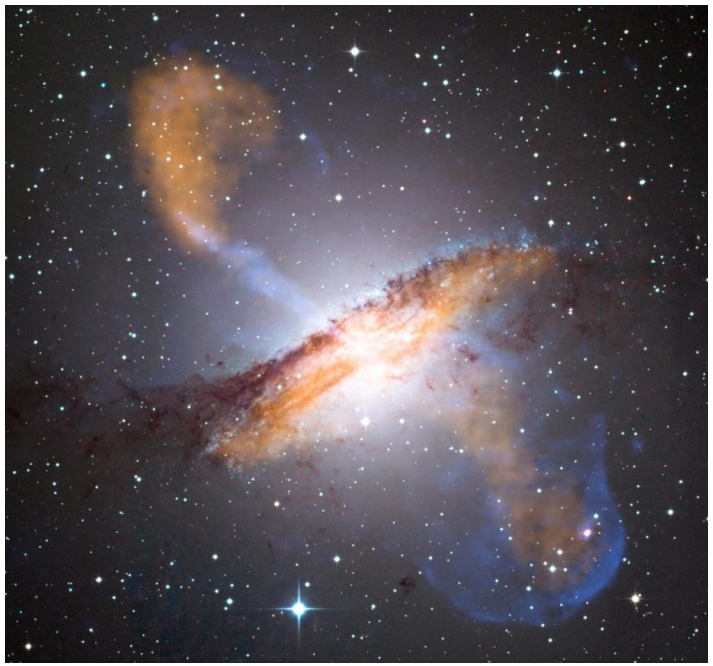


NIRSpec/PRISM spectrum
Synthetic NIRSpec photometry
NIRCam photometry (ForcePho)
NIRCam photometry (CIRC2)

A means to study high-redshift galaxies & AGN

- Galaxies produced the photons that drove reionization.
- Understanding reionization = understanding its cause(s)





Figures: Tilvi+20, George Mason University

Talk Outline

1. **How do we model cosmic reionization?**
2. **Timing of reionization - when did it start and end?**
3. **What can reionization tell us about high-z galaxies?**
4. **How did reionization affect IGM structure (& vice/versa)?**

Modeling Cosmic Reionization

Ingredients of a Successful Reionization Model

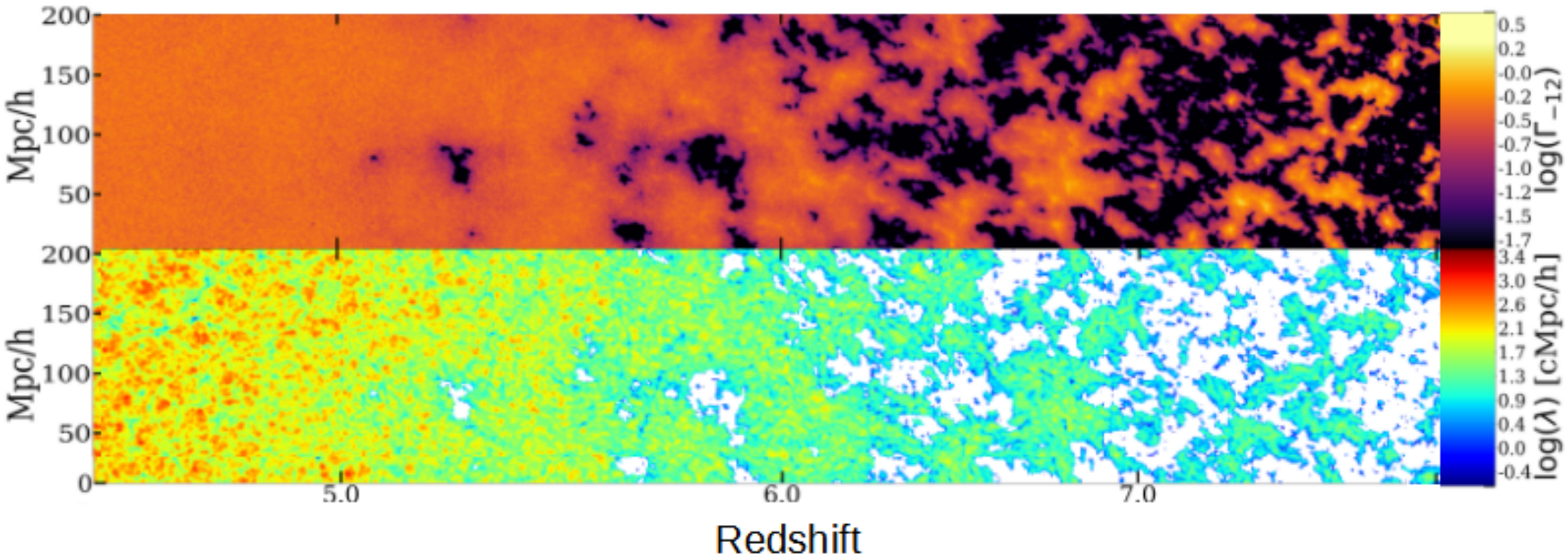
1. Ionizing photon sources (producers)
2. Radiative transfer (how photons travel)
3. Ionizing photon sinks (absorbers)

The Ionizing Sources

- Dark matter halos from a cosmological N-body simulation
- Halo mass function complete to $\sim 3 \times 10^9\ h^{-1} M_\odot$
- $M_{\rm UV}$ assigned by abundance matching (e.g. Adams+24)
- Mapping from $L_{\rm UV} \rightarrow \dot{n}_{\rm ion}$ determined by *free parameters*

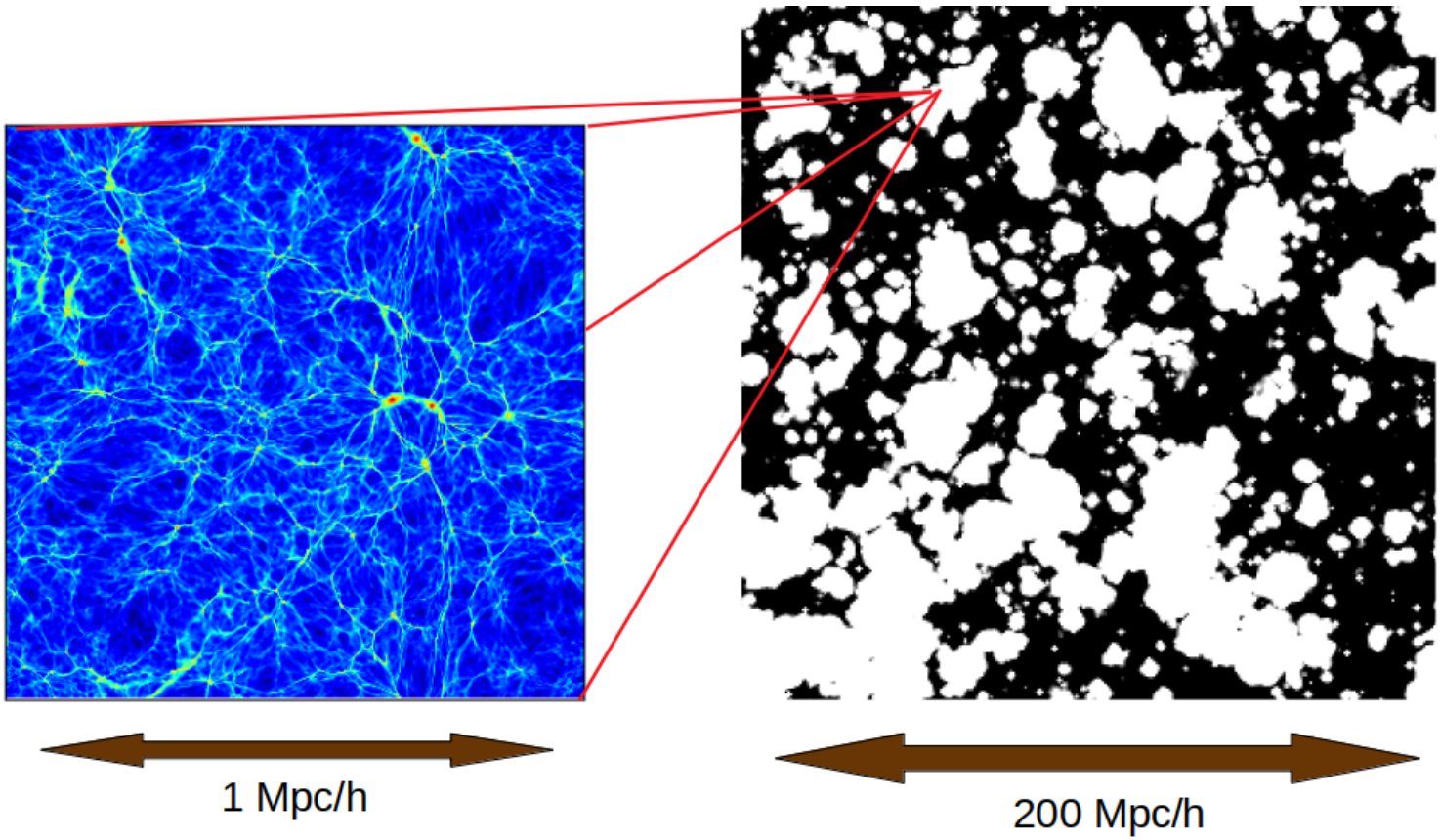
Radiative Transfer on large scales: FlexRT

- Ionizing radiative transfer in post-processing
- Adaptive ray tracing + large cell size = fast enough for parameter studies



Large-scale RT + small-scale IGM structure

- High-resolution IGM simulations in small boxes (1 h^{-1}cMpc)
- Novel RT method designed for sub-grid IGM opacity



Testing & Validation

Suite of simple post-processing RT validation tests from the Cosmological Radiative Transfer Code Comparison Project

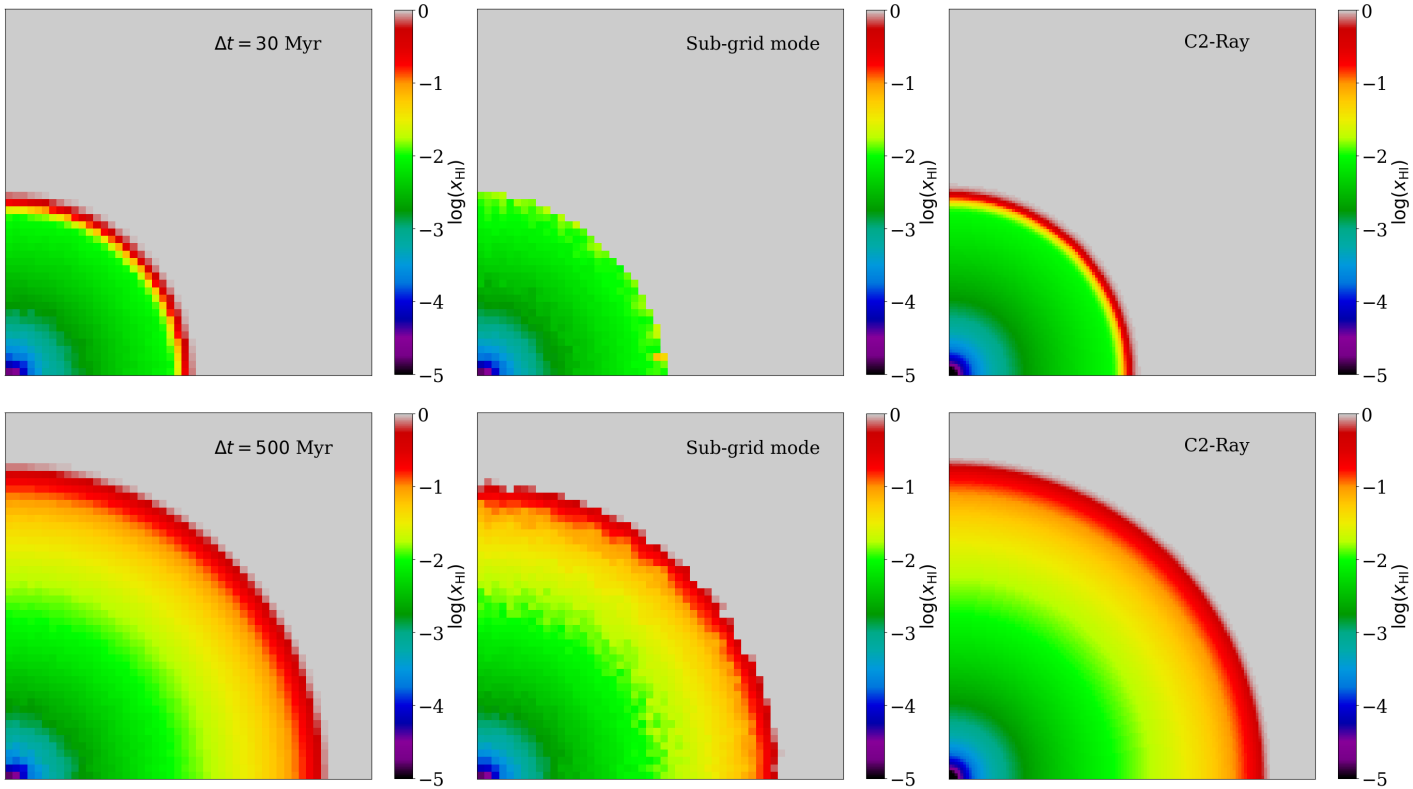
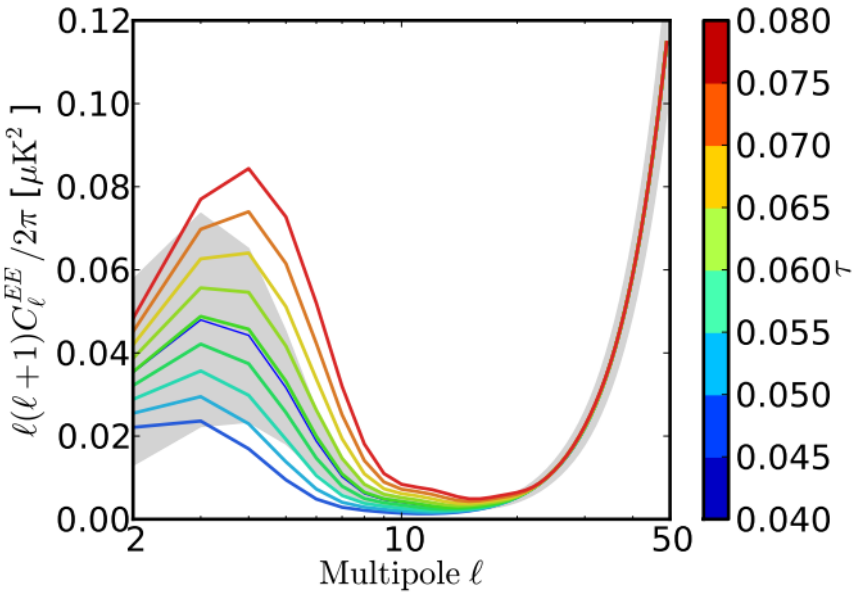


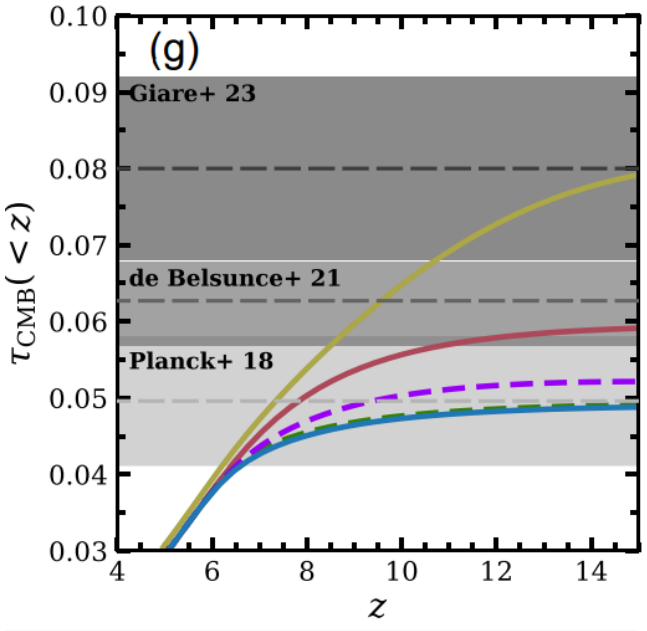
Figure: **CC**+24a

Timing of Reionization

Reionization's midpoint

- Constrained to $z_{\rm mid} \sim 7.5$ using Planck+18 measurements reionization optical depth (using EE mode power spectrum)
- More recent measurements suggest larger uncertainties

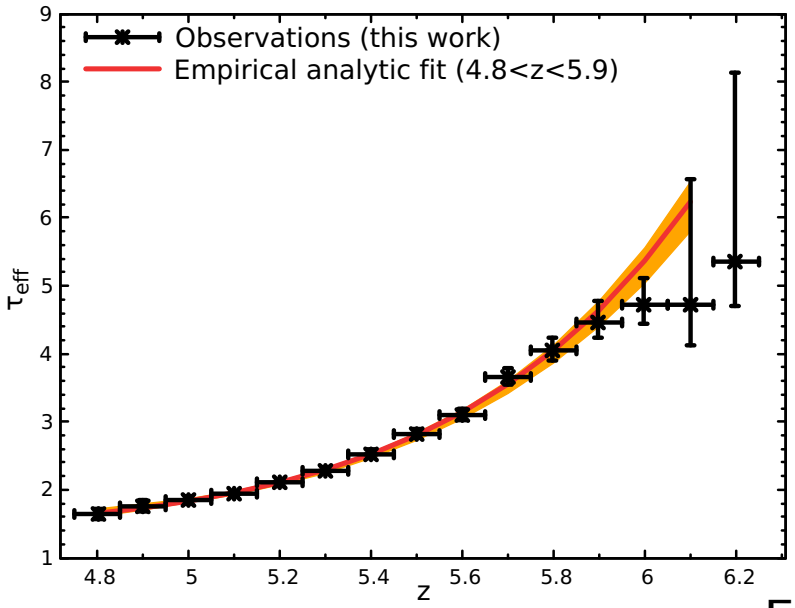


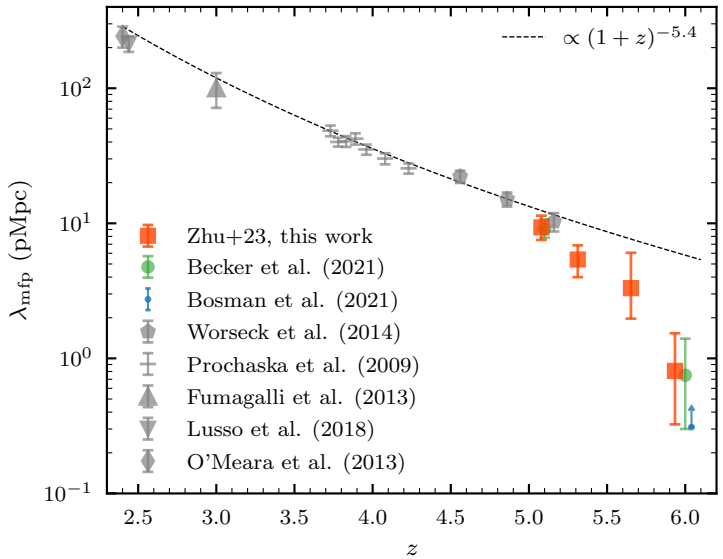


Figures: Planck+16, Asthana+24

What about the *end* of reionization?

- Thanks to high-redshift quasar spectra (XQR-30), we know when reionization *ended* very well
- Rapid evolution in the Lyα forest transmission and ionizing photon mean free path observed at $5 < z < 6$

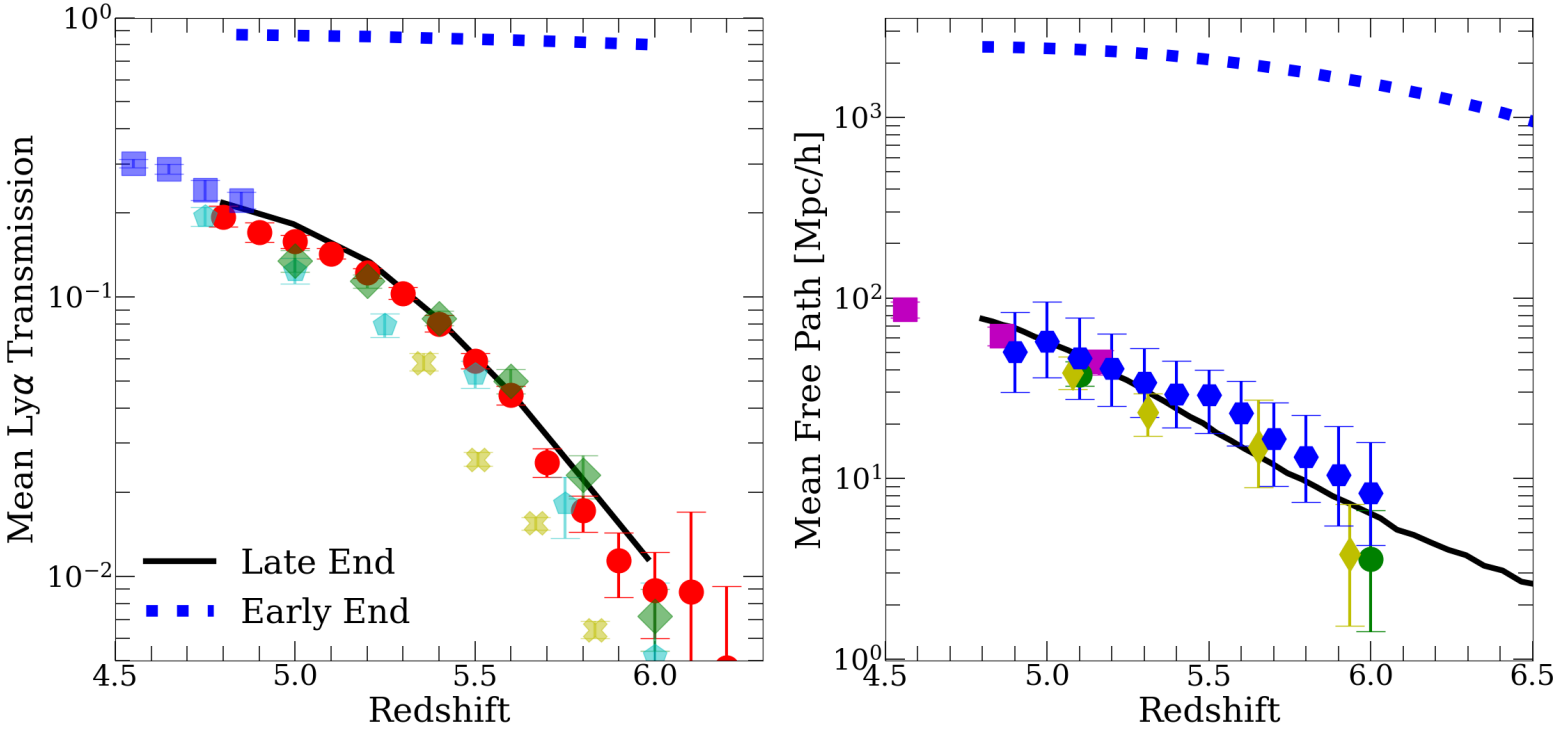




Figures: Bosman+22, Zhu,...,**CC**+23

Why reionization *cannot* end too early

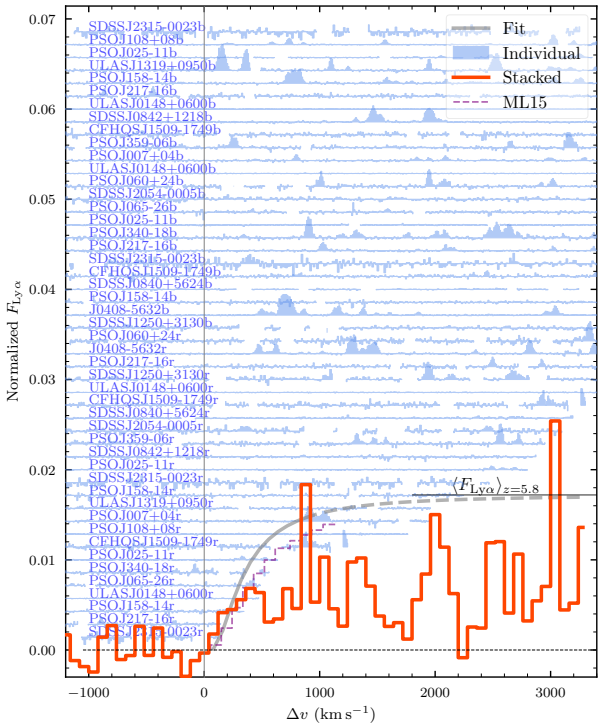
- The IGM is too transmission if reionization ends too early
- Several works place the end of reionization at $z = 5 - 5.5$ (e.g. Keating+19, Nasir+20, **CC+21, 23, 24b**, Qin+24)

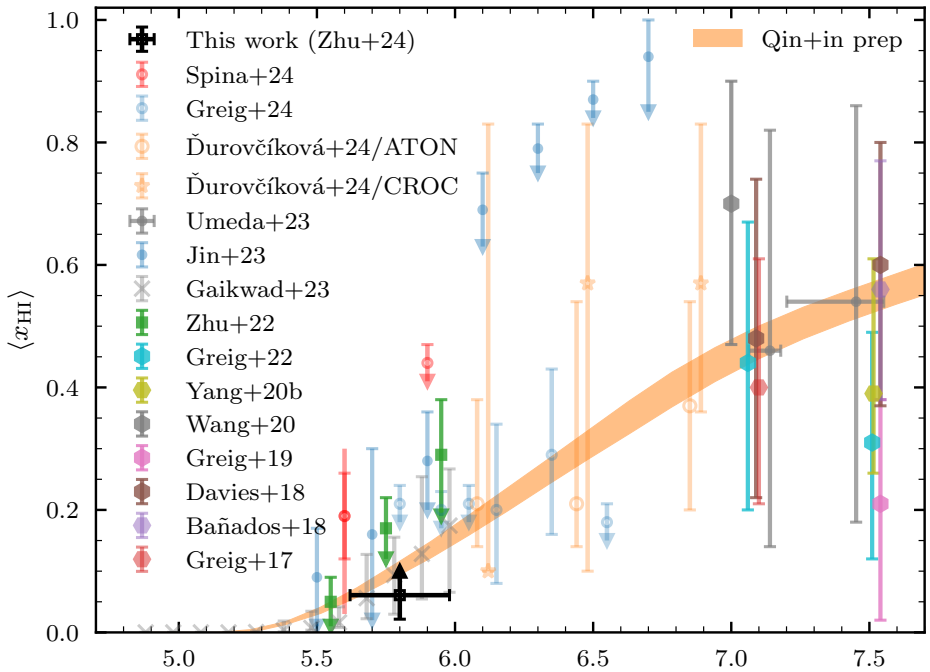


Figures: **CC**+24b

Have we *detected* late reionization?

- Probably, yes!
- Stacked Lyα forest spectra reveal damping wing-like features
- Constrain IGM to be at least 6% neutral at $z = 5.8$

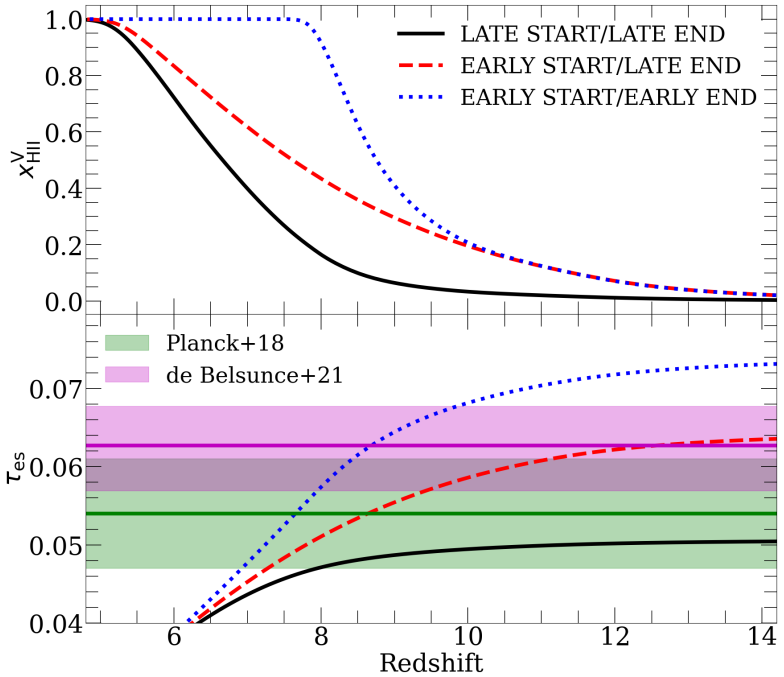


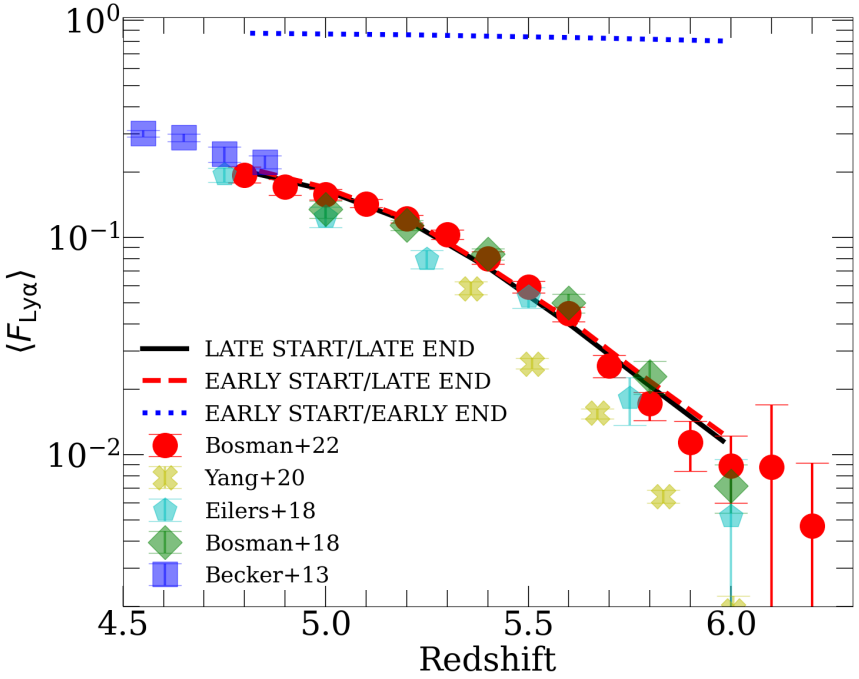


Figures: Zhu,...,**CC**+24

When did reionization *start*?

- This is harder...different reionization can start at very different times and still agree with the CMB and forest transmission

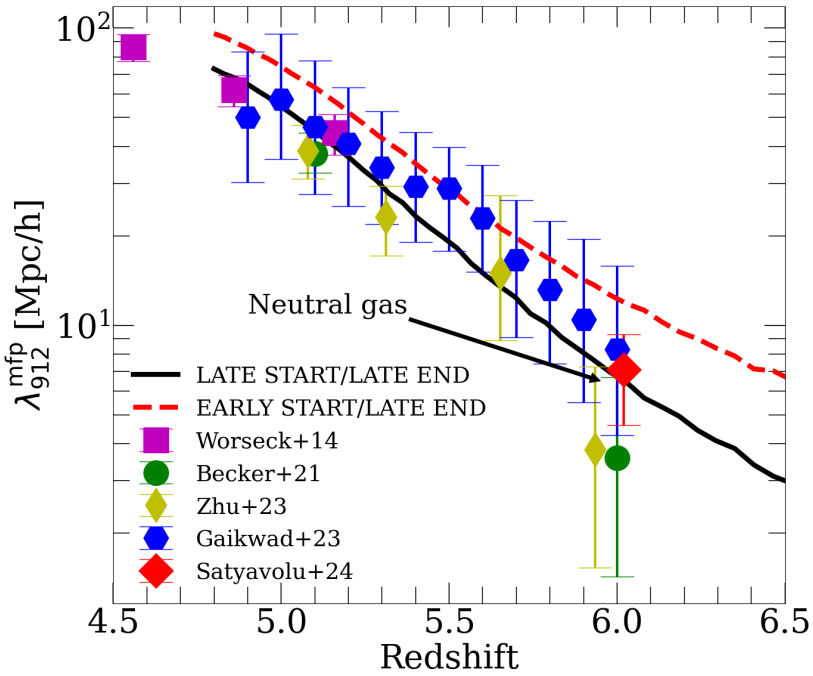


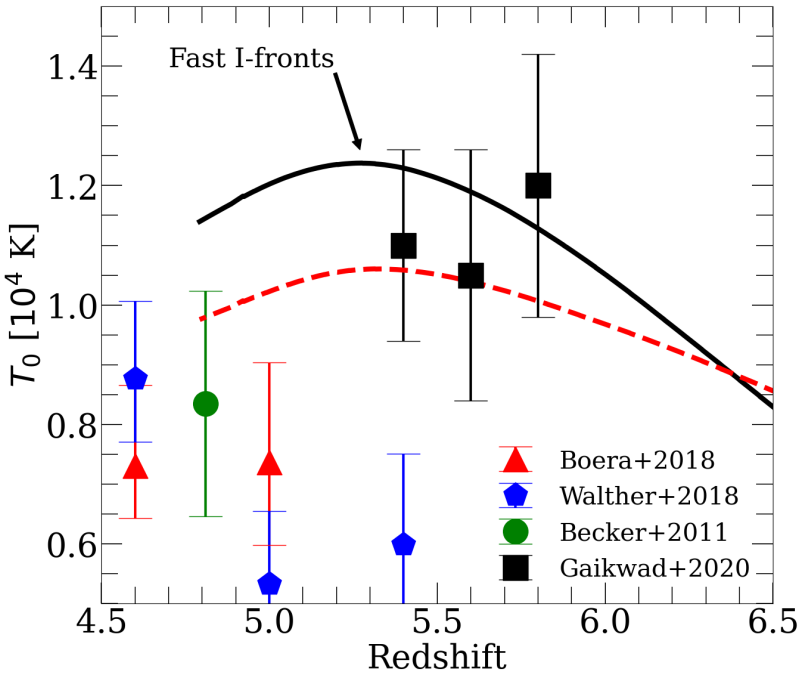


Figures: **CC**+24b

QSO Observations at $5 < z < 6$

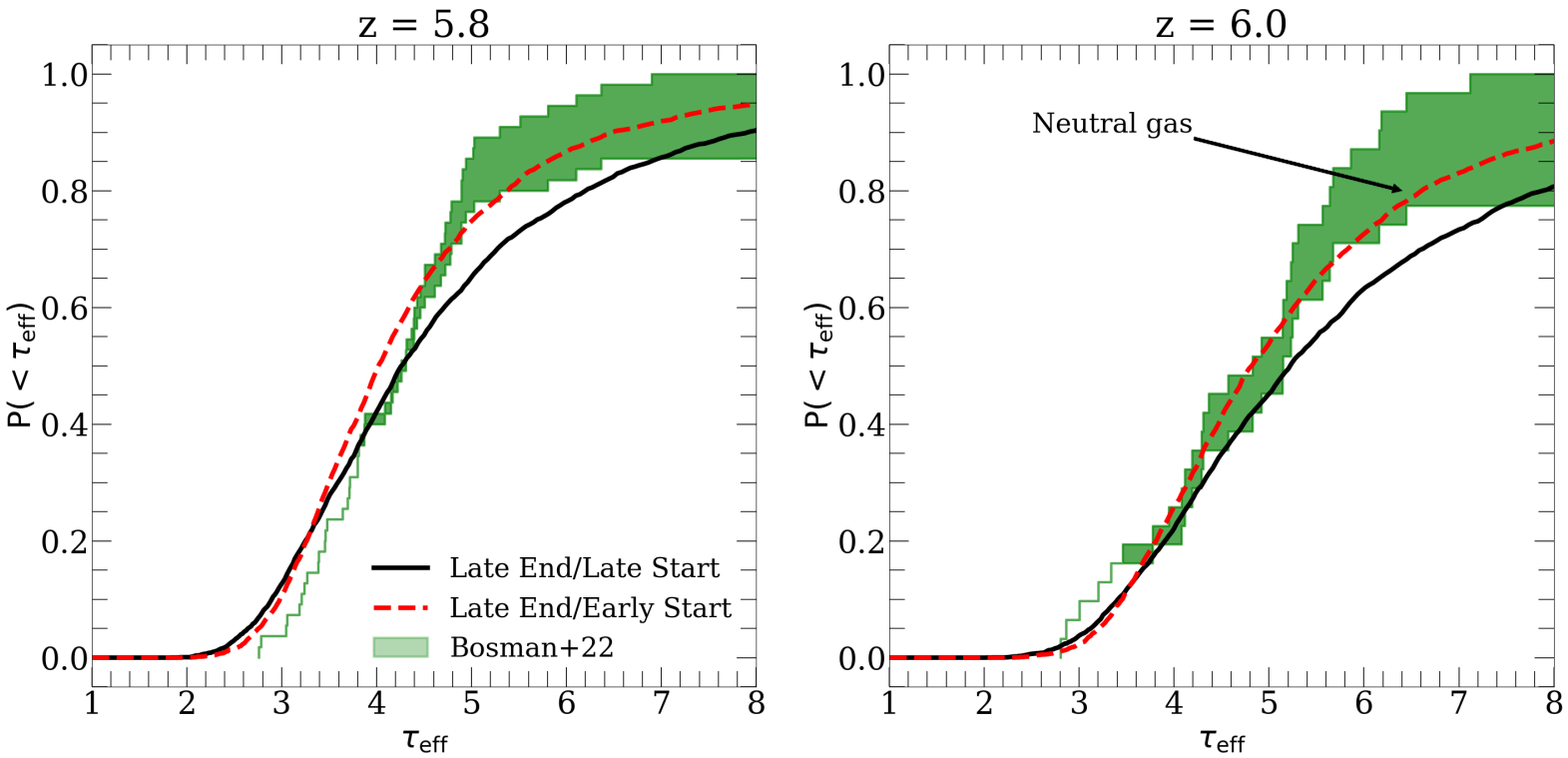
- The ionizing photon mean free path prefers a **late start**
- The thermal history of the IGM prefers an **early start**





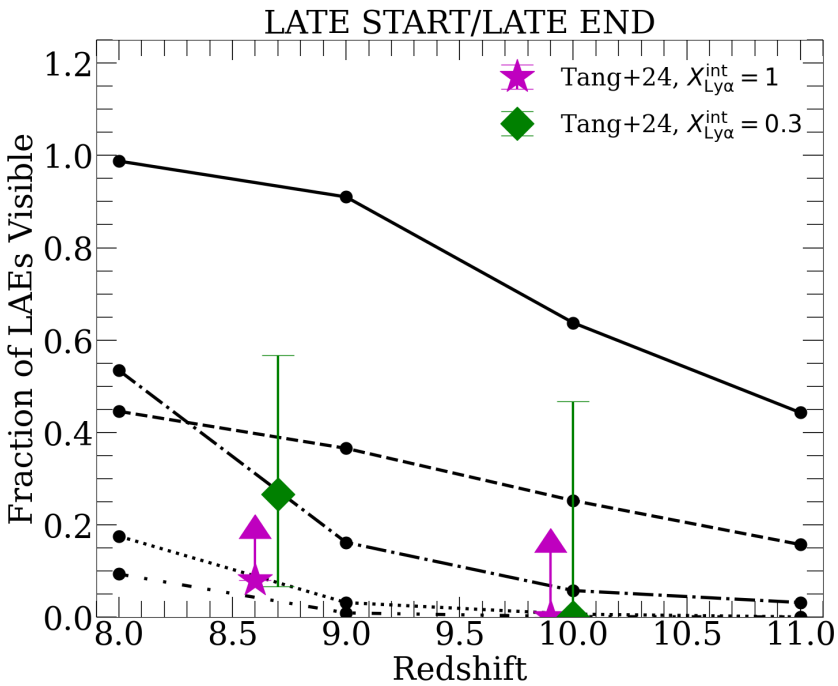
Scatter in the Lyα forest opacity

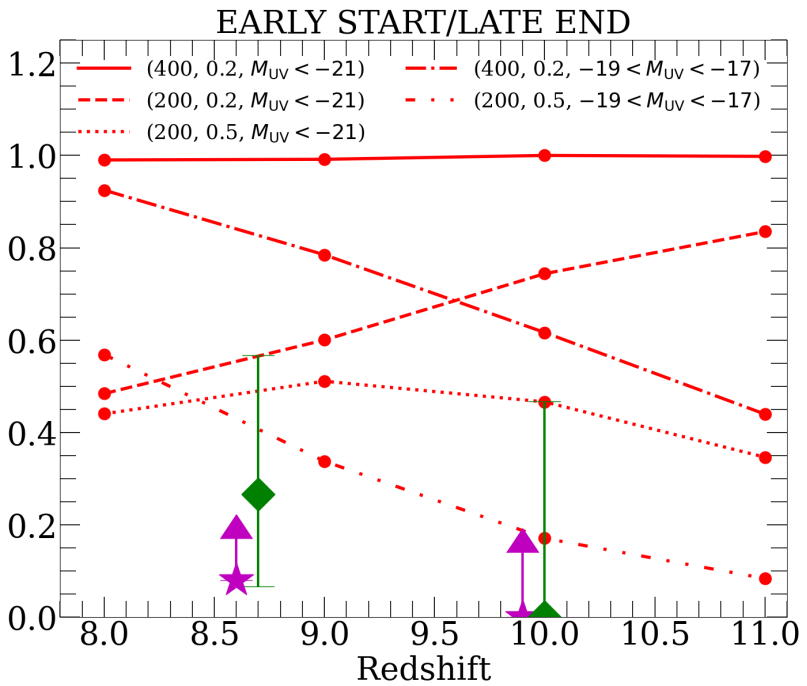
- Distribution of forest optical depths - sensitive to $x_{\rm HI}$
- An **early start** is preferred



How does Lyα emitter visibility evolve?

- Recent measurements (Tang+24) show sharp drop to $z \sim 10$
- We can estimate visibility in simulations for LAEs with different properties (detection criteria, velocity offset, $M_{\rm UV}$)
- Low visibility $\rightarrow$ **late start**

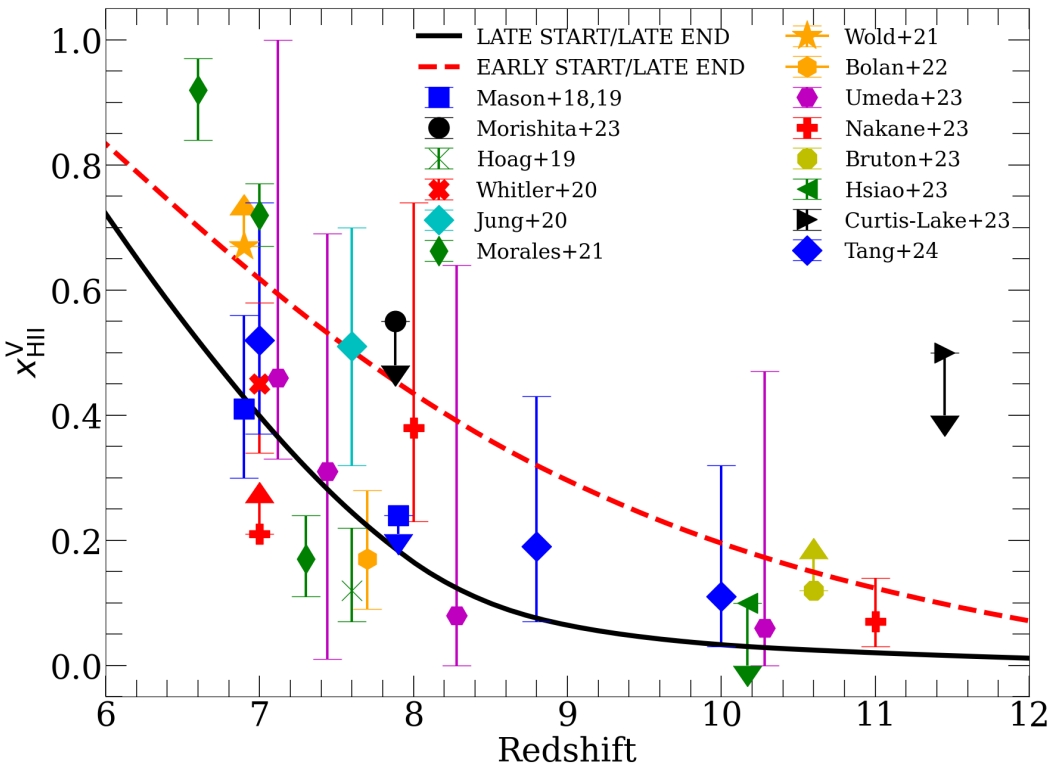




Measurements of $x_{\rm HI}$ at $z > 6.5$

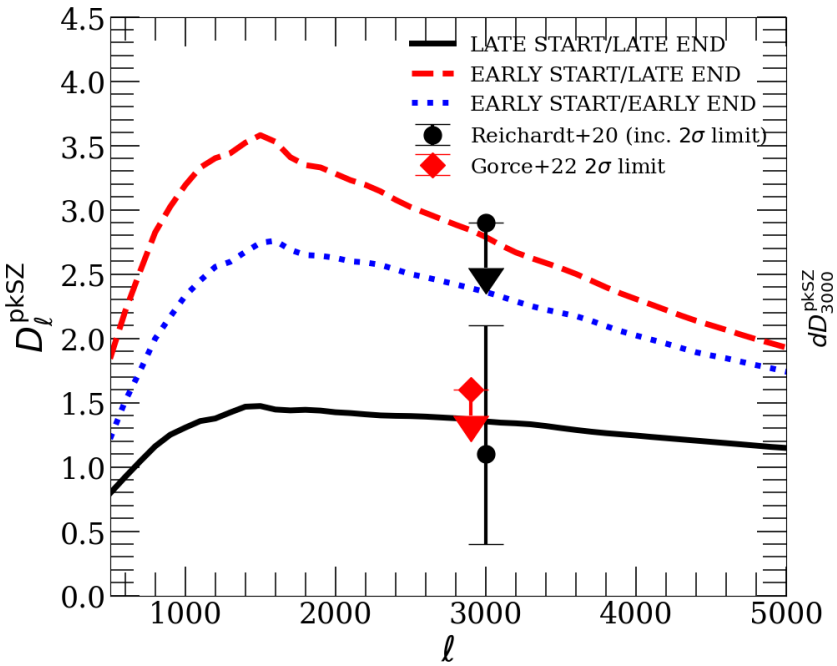
- Can infer $x_{\rm HI}$ with LAE detections and/or damping wings
- Measurements at $7 < z < 8$ do not favor late or early start
- Very few constraints at $z > 8$

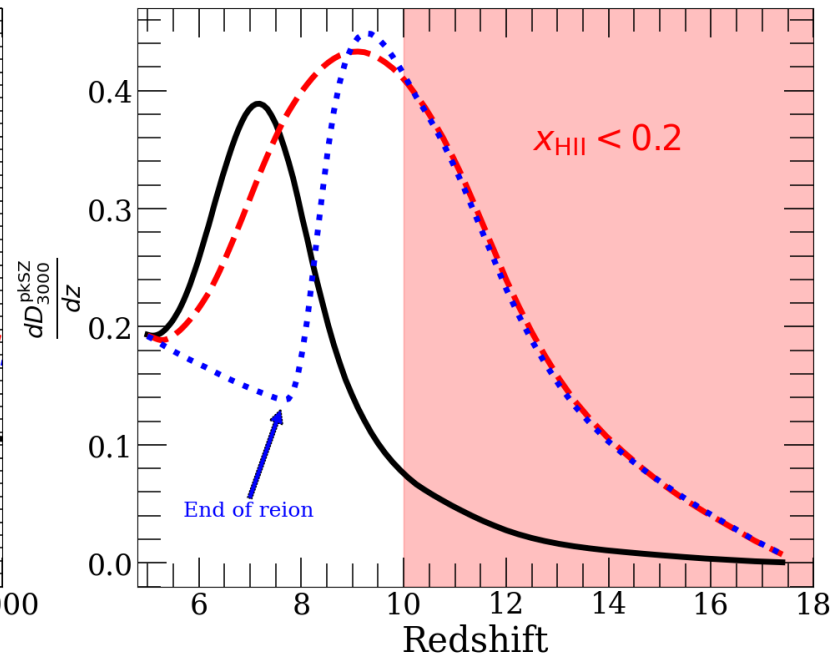
Inconclusive



Patchy kSZ effect from the CMB

- A **late start** is clearly favored by SPT measurement of the pkSZ effect (Reichardt+21)
- Much of the power in the early start case comes from reionization's beginning ($x_{\rm HI} < 0.2$)





Do observations prefer a late or early start?

Category	Probe	Late Start	Early Start
CMB	$\tau_{\rm es}$	No Pref.	No Pref.
	Patchy kSZ	Preferred	Not preferred
High-z Galaxies	UVLF/$\xi_{\rm ion}$/$f_{\rm esc}$	Preferred	Not preferred
	LAEs at $z > 8$	Preferred	Not preferred
	$x_{\rm HI}(z > 6.5)$	No Pref.	No Pref.
$z < 6.5$ QSOs	$\langle F_{\rm Ly\alpha} \rangle$	No Pref.	No Pref.
	$P(< \tau_{\rm eff}^{50})$	Not preferred	Preferred
	Mean Free Path	Preferred	Not preferred
	Thermal History	Not preferred	Preferred
	$x_{\rm HI}(z < 6.5)$	Not preferred	Preferred

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	Mean Free Path	Preferred	Not preferred
	Thermal History	Not preferred	Preferred
	$x_{\rm HI}(z < 6.5)$	Not preferred	Preferred
Final Score	All Data	Preferred	Not preferred

Timing of Reionization: Summary

- The end of reionization is well-constrained by high-redshift quasar data to $5 \lesssim z \lesssim 5.5$
- The start of reionization is poorly constrained, and different data sets may be giving different answers
- More and better data will clarify our picture of when reionization started (and/or reveal systematics in observations & models!)

Probing High-z Galaxies with Reionization

What can reionization tell us about galaxies?

- Which sources did reionization? Stars or AGN? (D'Aloisio+17, Garaldi+18, Madau+24, Jiang+25)
- **Collective ionizing output** (Becker+13, Gaikwad+23, **CC**+25)
- Clustering properties (Naidu+20, **CC**+22b)
- Feedback processes in e.g. low-mass galaxies (Wu+19, Ocvirk+21, **CC**+23)

The ionizing emissivity at reionization's end

- Simulations that match $5 < z < 6$ Lyα forest measurements require a drop in ionizing output at $z \lesssim 6$
- This is puzzling because the UV luminosity density of galaxies rises steadily over this period
- Is this galaxy physics, or an artifact of inaccurate modeling?

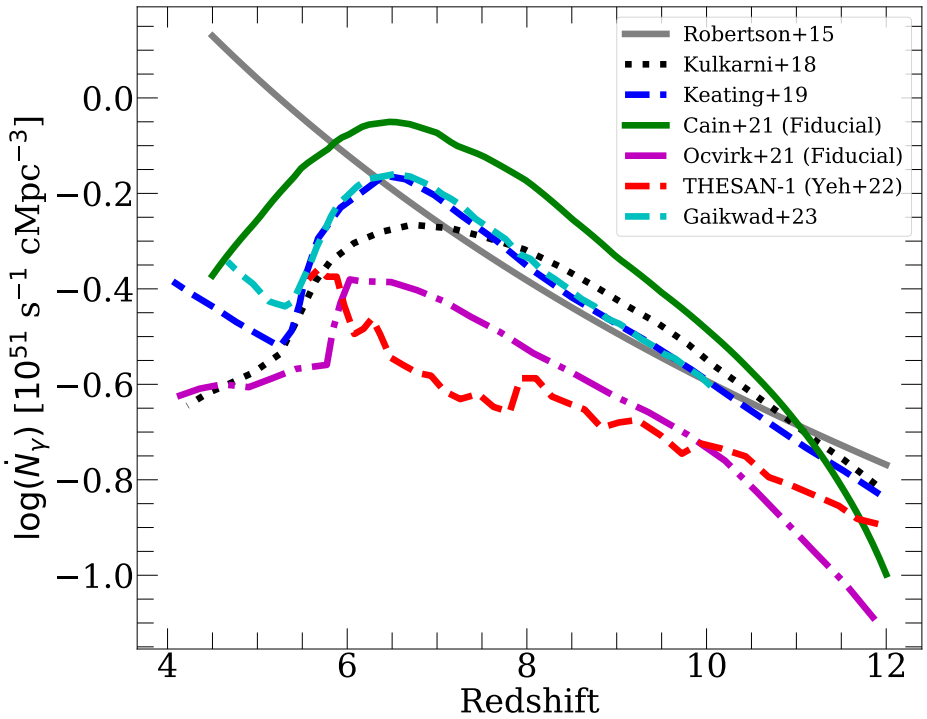
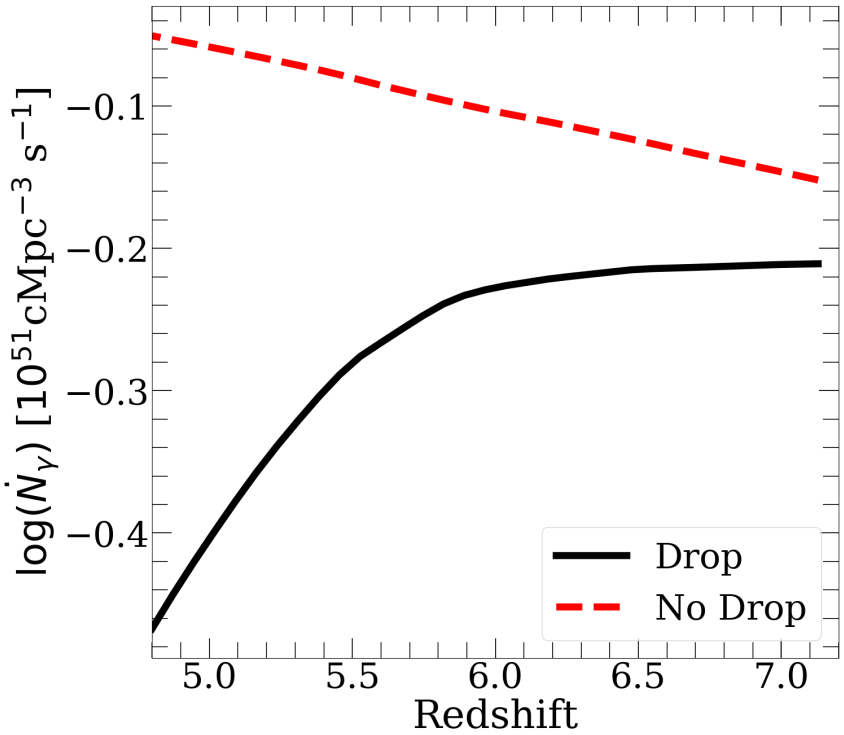
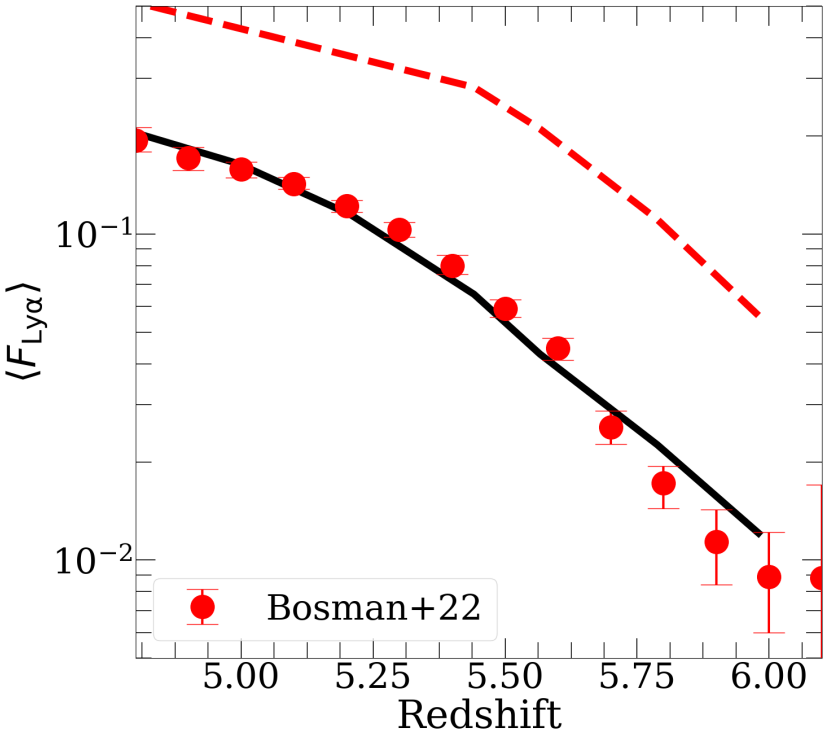


Figure: **CC**+23

Why is the drop needed?

- Without it, models tend to overshoot the $5 < z < 6$ forest



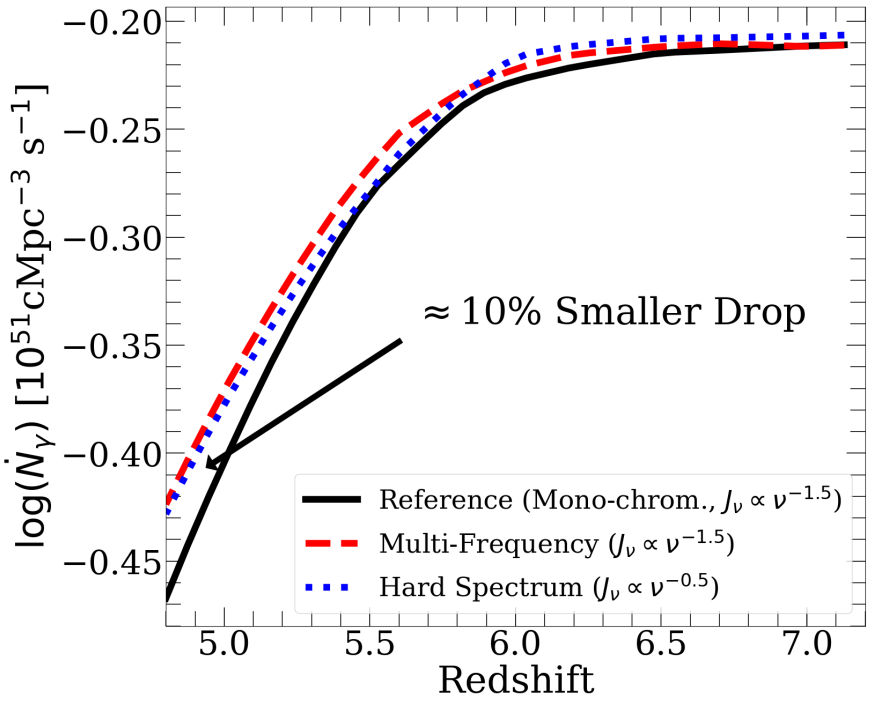


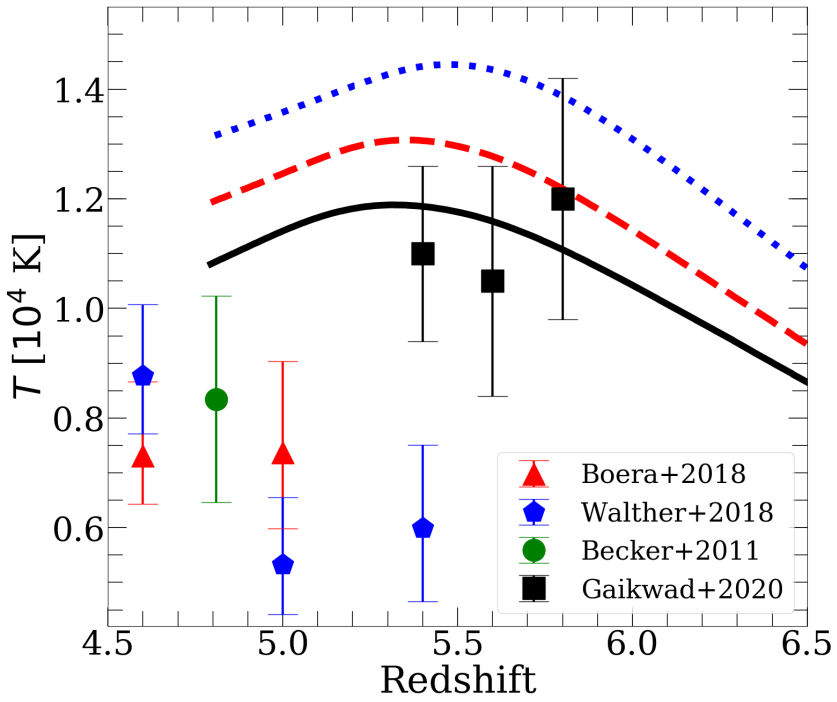
What could we be missing in the IGM?

- Thermal history of the IGM? (Keating+19)
- Missing spatial resolution? (un-resolved sinks)
- Rare, massive sinks? (Munoz+15, **CC+21**)
- Something else?

Ionizing Spectrum (for fixed mean forest flux vs. z)

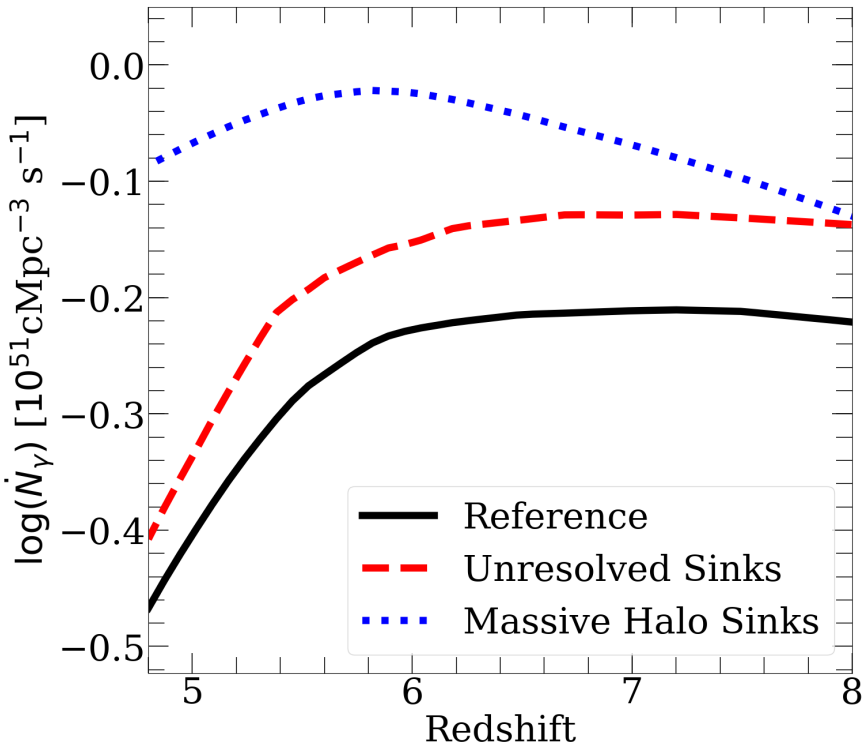
- Harder ionizing spectra lower $\Gamma_{\rm HI}$ but raise T
- Small effect on $\dot{N}_\gamma$ drop, limited by T measurements

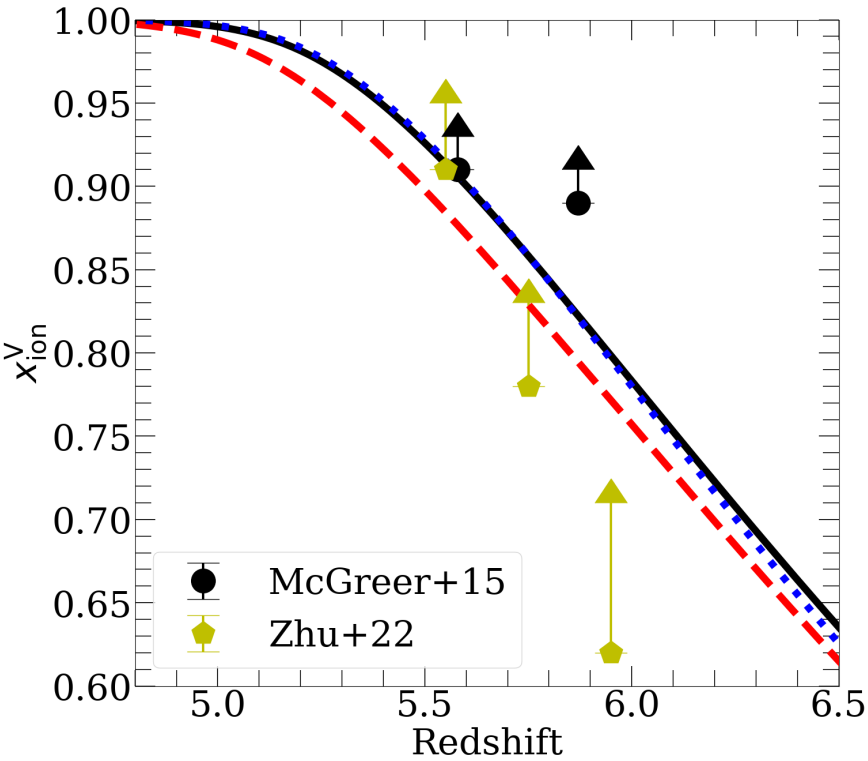




Missing Sinks (for fixed mean forest flux vs. z)

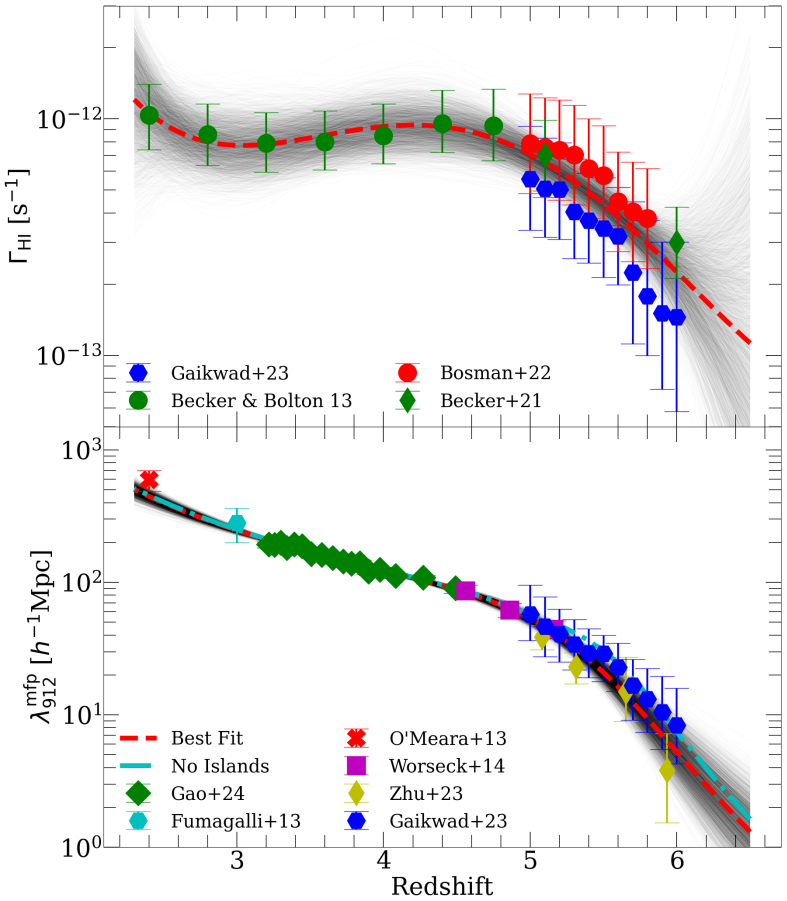
- Extra rare, massive sinks significantly reduce the drop
- **Forest observations likely require at least some evolution in galaxy ionizing properties**





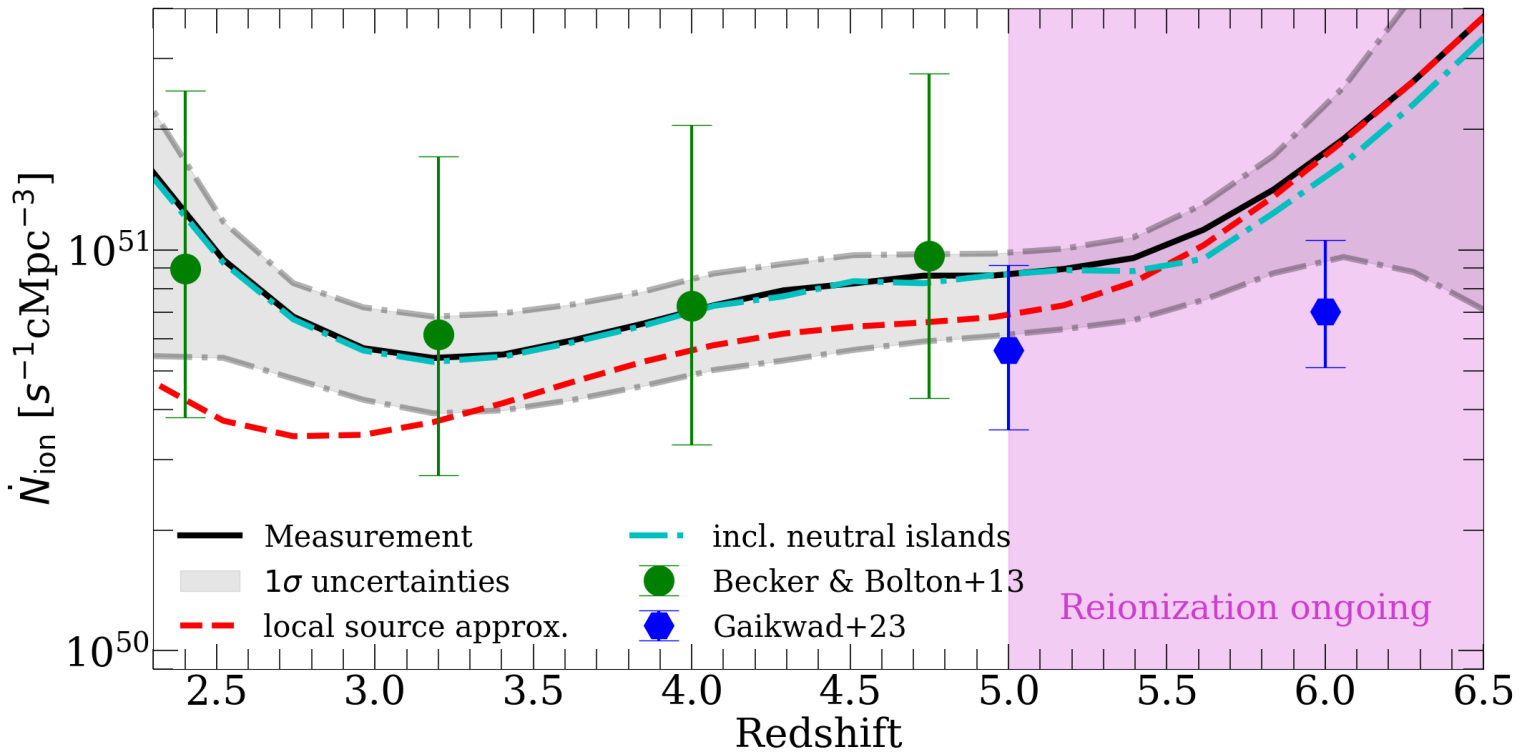
Can we *measure* the ionizing output of galaxies?

- Yes we can...using quasar spectra!
- Measurements of the HI photo-ionization rate ($\Gamma_{\rm HI}$) and mean free path ($\lambda_{912}^{\rm mfp}$) can be combined to measure the ionizing emissivity $\dot{N}_{\rm ion}$
- Requires radiative transfer and accounting for neutral gas at reionization's end
- Measurements are available up to $z = 6$ (Figure: **CC**+25)



Fiducial Measurement

- $\dot{N}_{\rm ion}$ is flat at $2.5 < z < 5$, but starts increasing at $z > 5$
- Local source approximation: $\approx 20\%$ underestimate at $z = 5$



Measuring galaxy ionizing properties

Combining high-z quasar results with data from HST & JWST, we can measure the ionizing properties of the galaxy population

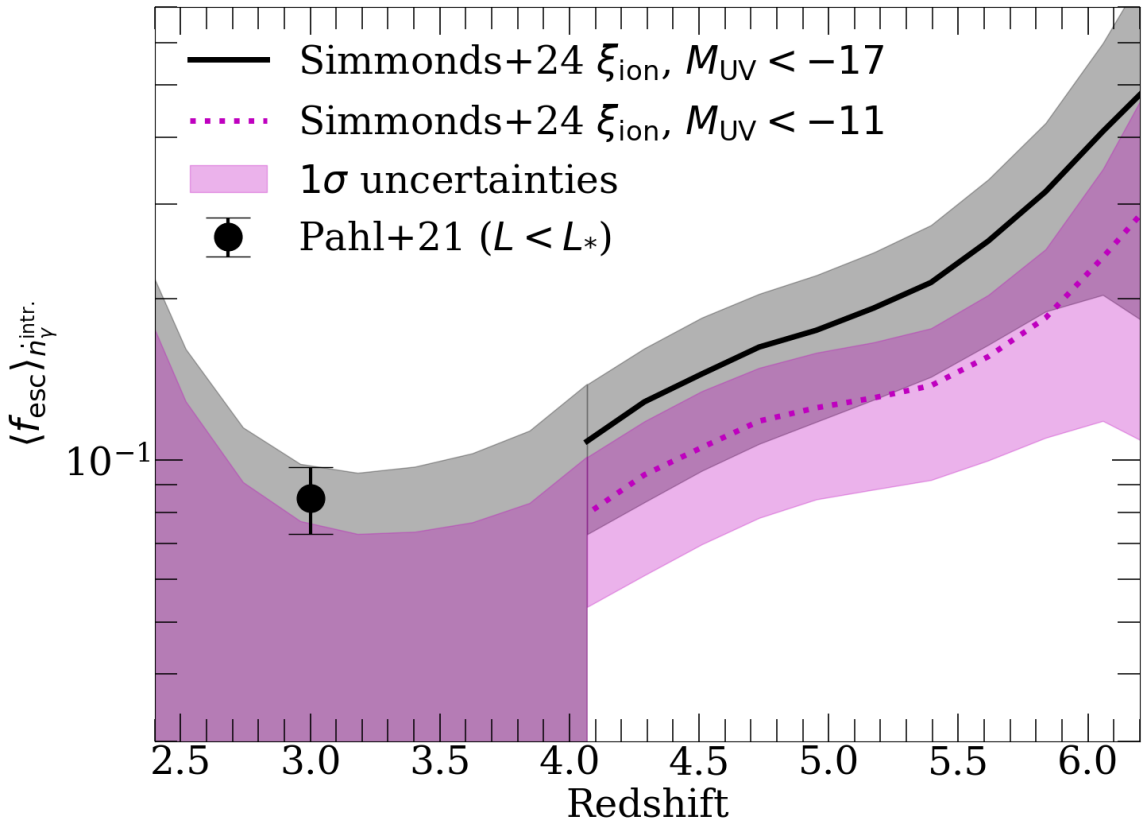
Escaping ionizing efficiency: $\langle f_{\rm esc}\xi_{\rm ion}\rangle_{L_{\rm UV}} = \frac{\dot{N}_{\rm ion}}{\rho_{\rm UV}}$

Average escape fraction: $\langle f_{\rm esc}\rangle = \frac{\dot{N}_{\rm ion}}{\dot{N}_{\rm ion}^{\rm intr.}}$

$$\dot{N}_{\rm ion}^{\rm intr.} = \int_{-\infty}^{M_{\rm UV}^{\rm cut}} dM_{\rm UV} \frac{dn}{dM_{\rm UV}} L_{\rm UV}\xi_{\rm ion}(z, M_{\rm UV})$$

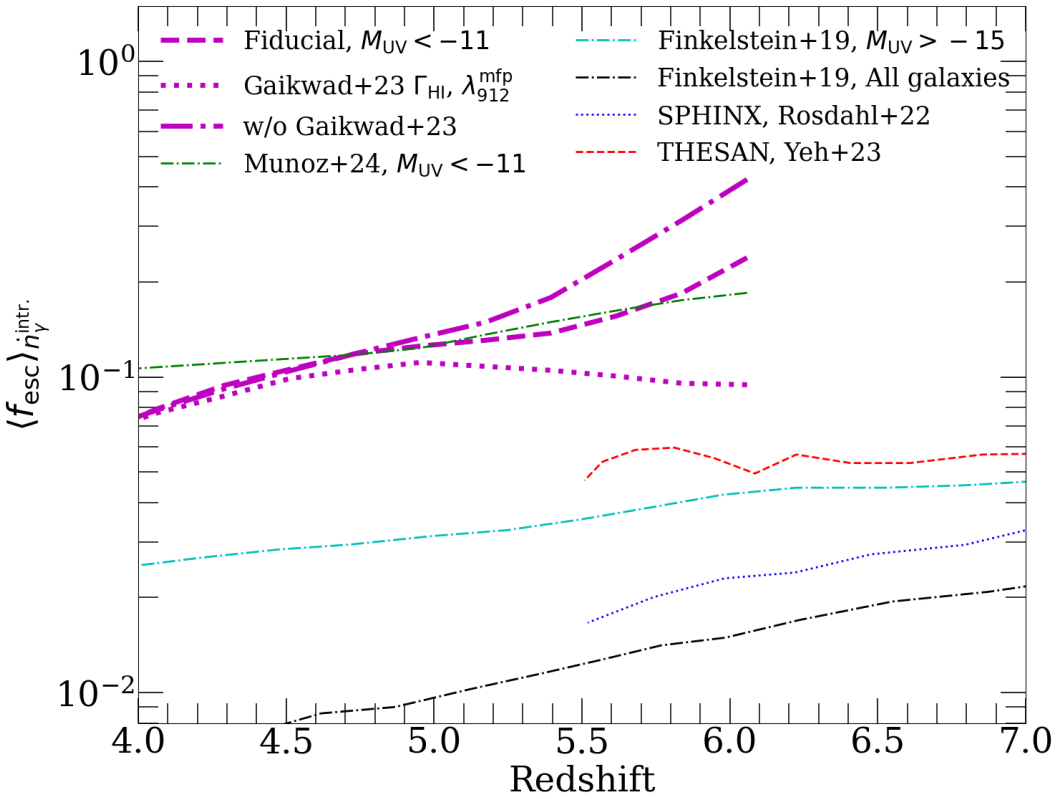
Fiducial measurements of $\langle f_{\rm esc} \rangle$

- Consistent with $f_{\rm esc} \lesssim 10\%$ at $z \lesssim 4$
- Mild increase between $z = 4 - 5$ to $10 - 20\%$
- Factor of ≈ 2 increase from $z = 5 - 6$



Indirect inferences of $f_{\rm esc}$

- We can compare our measurements to simulations and inferences based on low-z measurements
- Reasonable agreem with $\beta_{\rm UV}$-$f_{\rm esc}$ relation, but only for $M_{\rm UV} < -11$
- Simulations may systematically under-estimate $f_{\rm esc}$



Reionization & High-z Galaxies: Summary

- High-z quasar spectra are already giving us key insights into the ionizing properties of galaxies at reionization's end
- Simulations and measurements hint at rapid evolution in galaxy ionizing properties approaching $z = 6$ - but still with large observational and modeling uncertainties
- Forthcoming observations probing reionization will continue to refine our understanding up to $z = 6$ and beyond

Reionization & IGM Structure

Small-scale structure in the re-ionizing IGM

- Reionization impulsively heats the IGM by $2-3$ OoM
- Reionization evaporates "mini-halos" and pressure-smooths filaments, dramatically changing IGM conditions

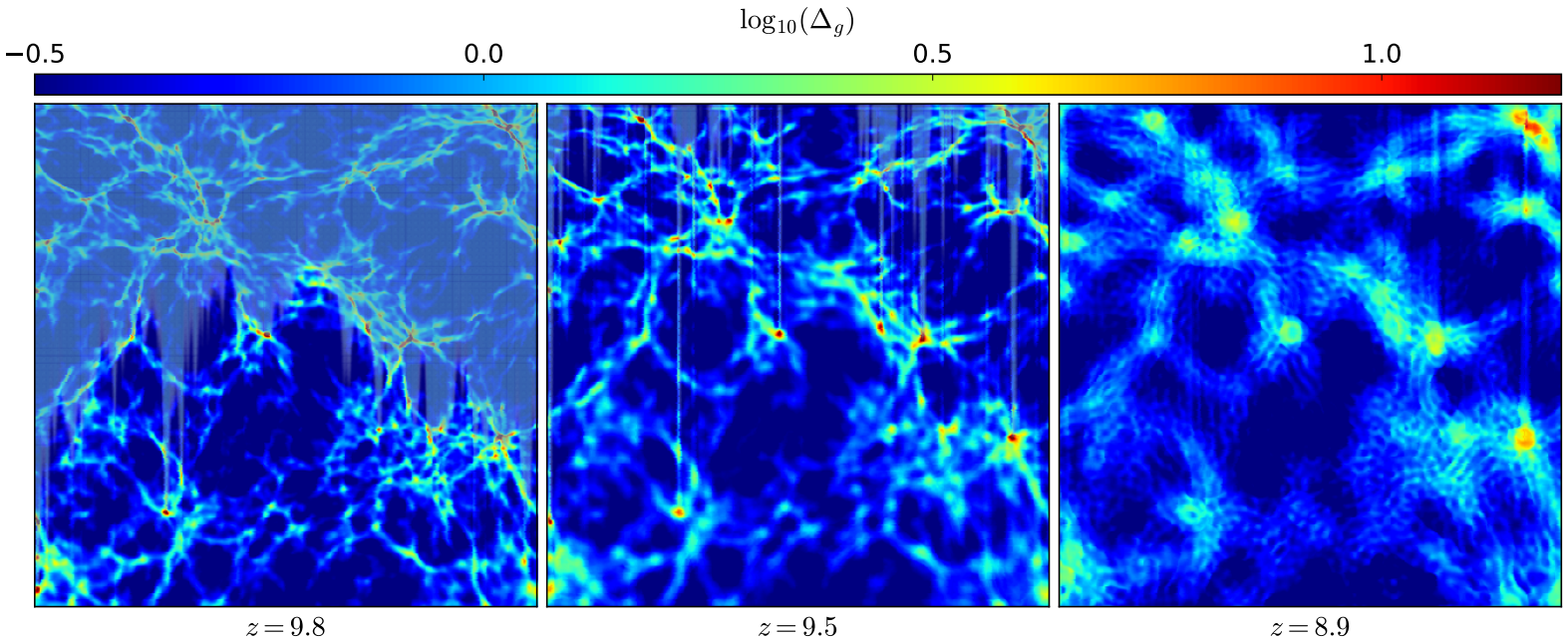


Figure: D'Aloisio,...,**CC**+20

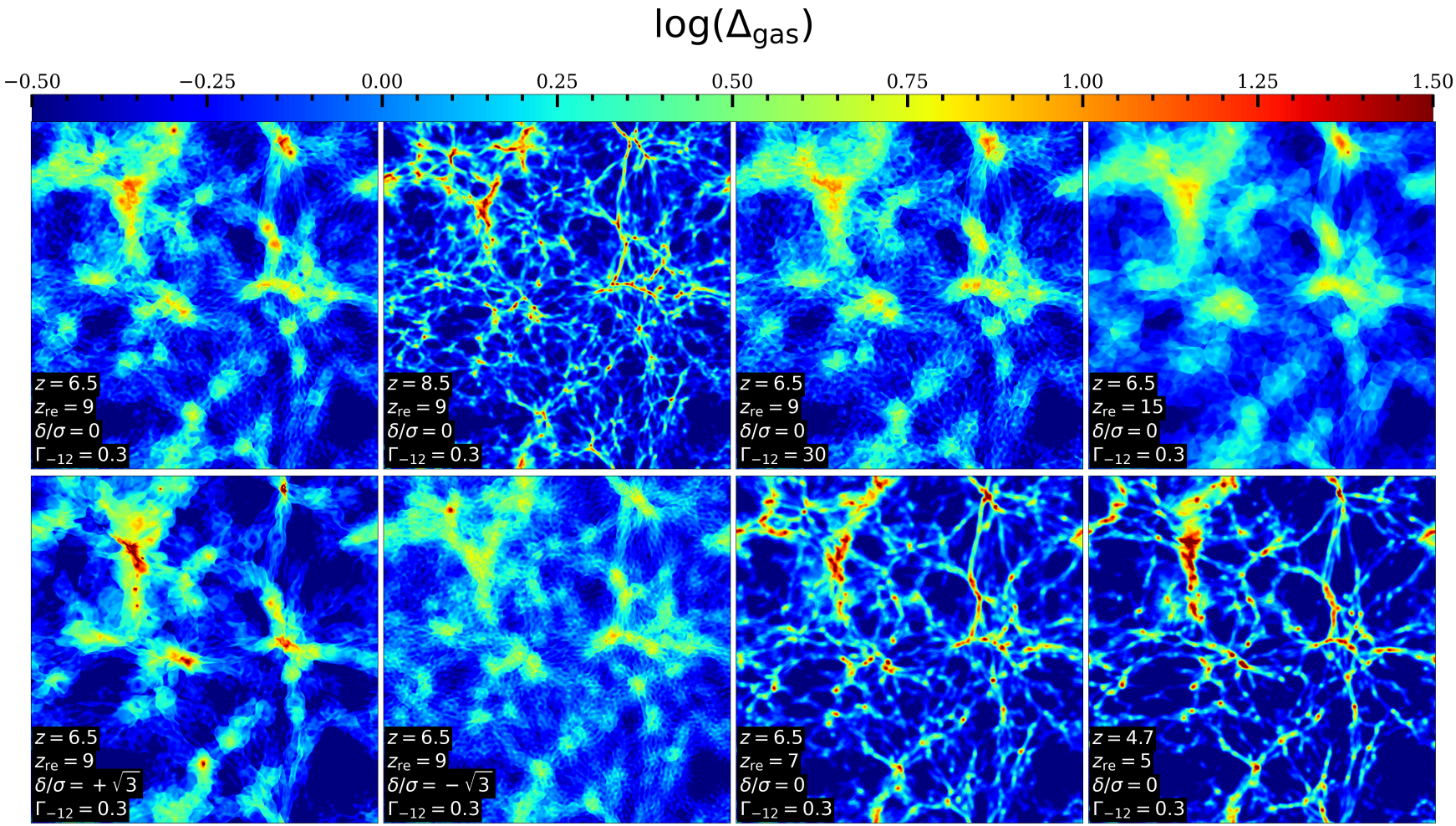
Effects of "IGM relaxation" on small scales

- Photo-evaporation of mini-halos with masses $\lesssim 10^8\ M_\odot$ (Shapiro+04, Iliev+05, Chan+23)
- Destruction of "Lyman-limit" HI absorption systems that set the IGM mean free path (Park+16, D'Aloisio,..,**CC**+20, Nasir,**CC**+21,**CC**+21, 22a)
- Back-reaction on the reionization photon budget and morphology (McQuinn+06, Davies+21, **CC**+21, 22b)
- **Alters the IGM thermal structure (Hirata+18, CC+24c)**
- **Drives turbulence in the IGM (CC+ in prep.)**

Improvements in IGM modeling

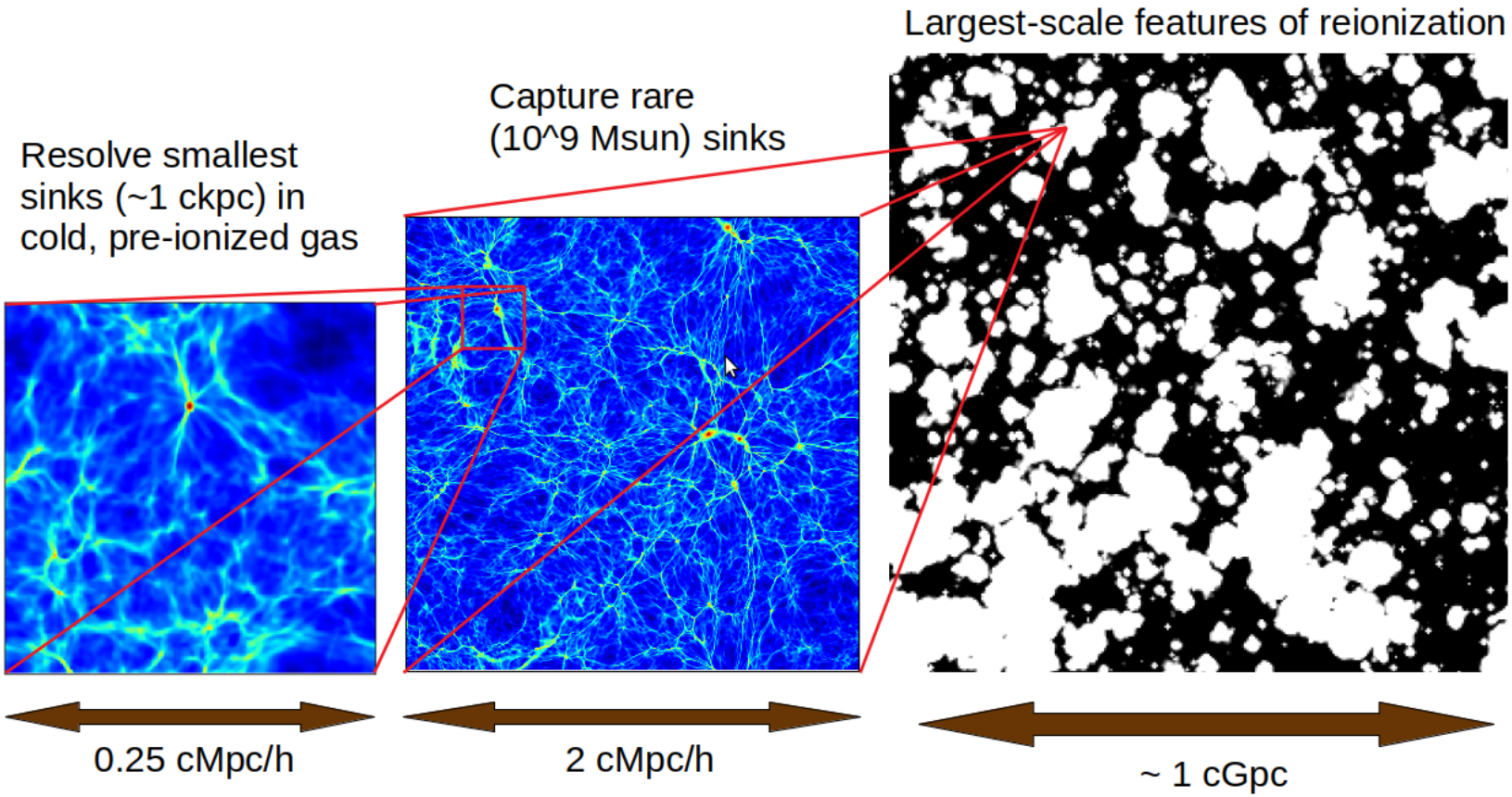
- Introducing **SAGUARO** (**S**imul**a**ting **IG**M Evol**u**tion and Environments **a**t High **R**es**o**lution)
- Suite of high-resolution coupled RT/hydro simulations of the IGM using RadHydro (Trac & Cen 2004, Trac+07)
- Spans $4 < z < 15$ (useful for post-reionization studies)
- Large-scale IGM environments: $\Gamma_{\rm HI}$ (ionizing background), $\Delta_{\rm box}$ (box-scale density), $z_{\rm reion}$ (redshift of reionization)
- 2 h^{-1}kpc (250 h^{-1}pc) resolution for $L_{\rm box} = 2$ h^{-1}Mpc (250 h^{-1} kpc)
- Phase 1: 64 simulations, $\sim$ 1.5 million CPU hours

Small-scale IGM dynamics across large-scale environments



Multi-scale sub-grid modeling

The Goal: Parameter space studies in ~ 1 Gpc boxes with RT simulations



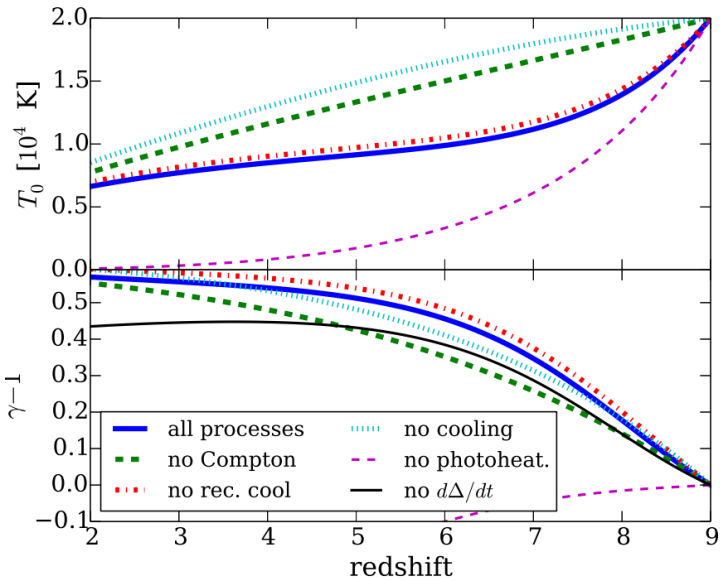
Thermal Structure of the IGM

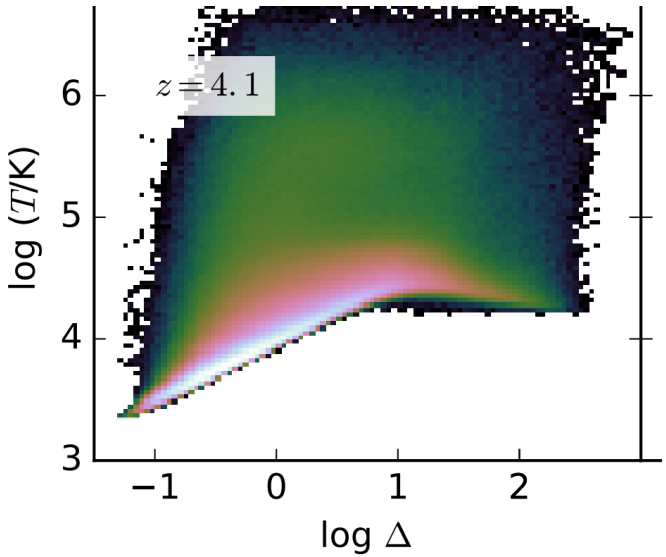
- Evolution of the mean IGM temperature and its fluctuations during and after reionization
- A key ingredient in forward-modeling the Lyα forest
- Important for: reionization timeline, measurements of $\Gamma_{\rm HI}$, constraints on dark matter models, etc.

The IGM Temperature-Density Relation

- The low-density IGM is believed to follow a tight power law temperature-density relation (TDR)

$$T(\Delta) = T_0 \Delta^{\gamma - 1}$$





Figures: McQuinn+15, Keating+17

Resolution convergence of the TDR

- $L_{\rm box} = 2\ h^{-1}$cMpc with 8 h^{-1}ckpc spatial resolution - on the high-end for standard Lyα forest simulations

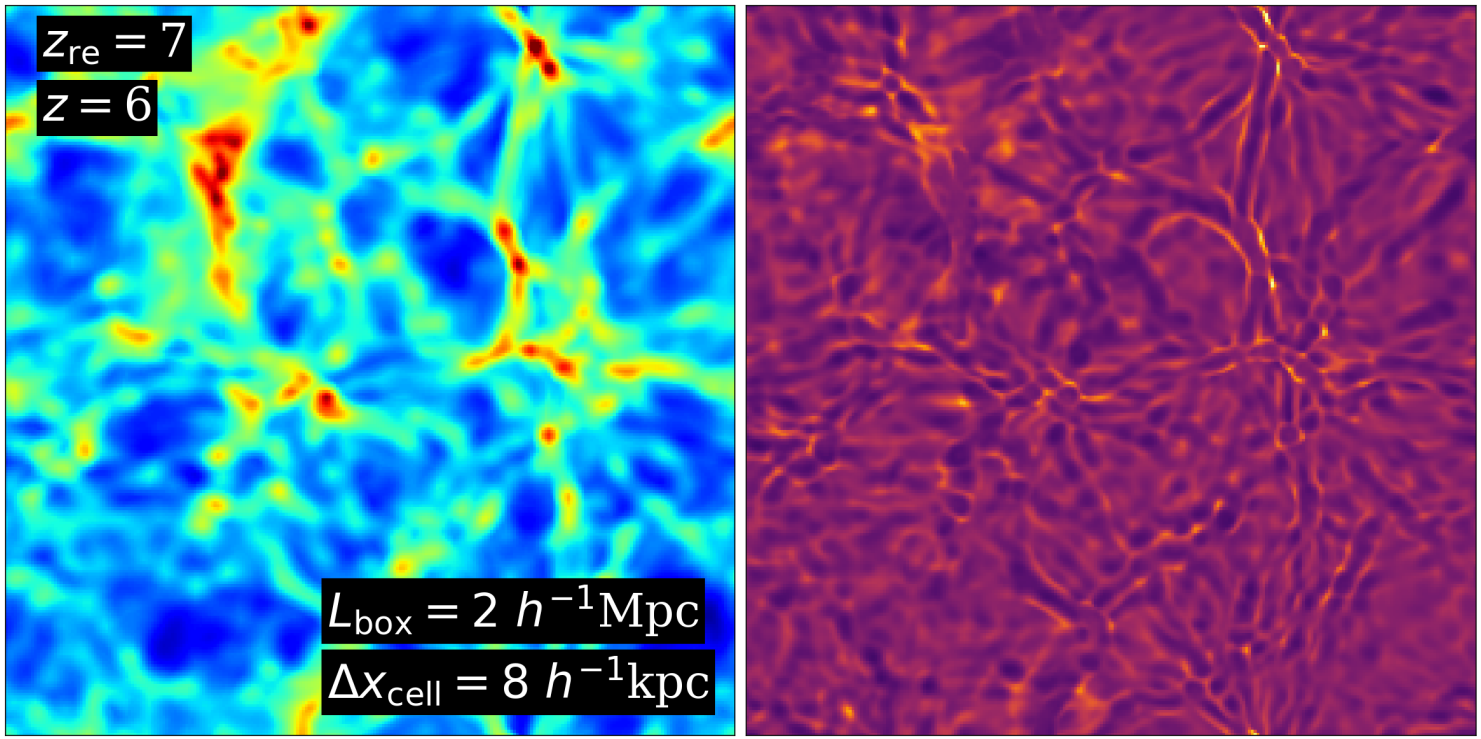


Figure: **CC**+24c

Resolution convergence of the TDR

- Increasing resolution to 2 h^{-1}ckpc (by a factor of $4^3\times$) makes a significant difference for small-scale thermal structure!

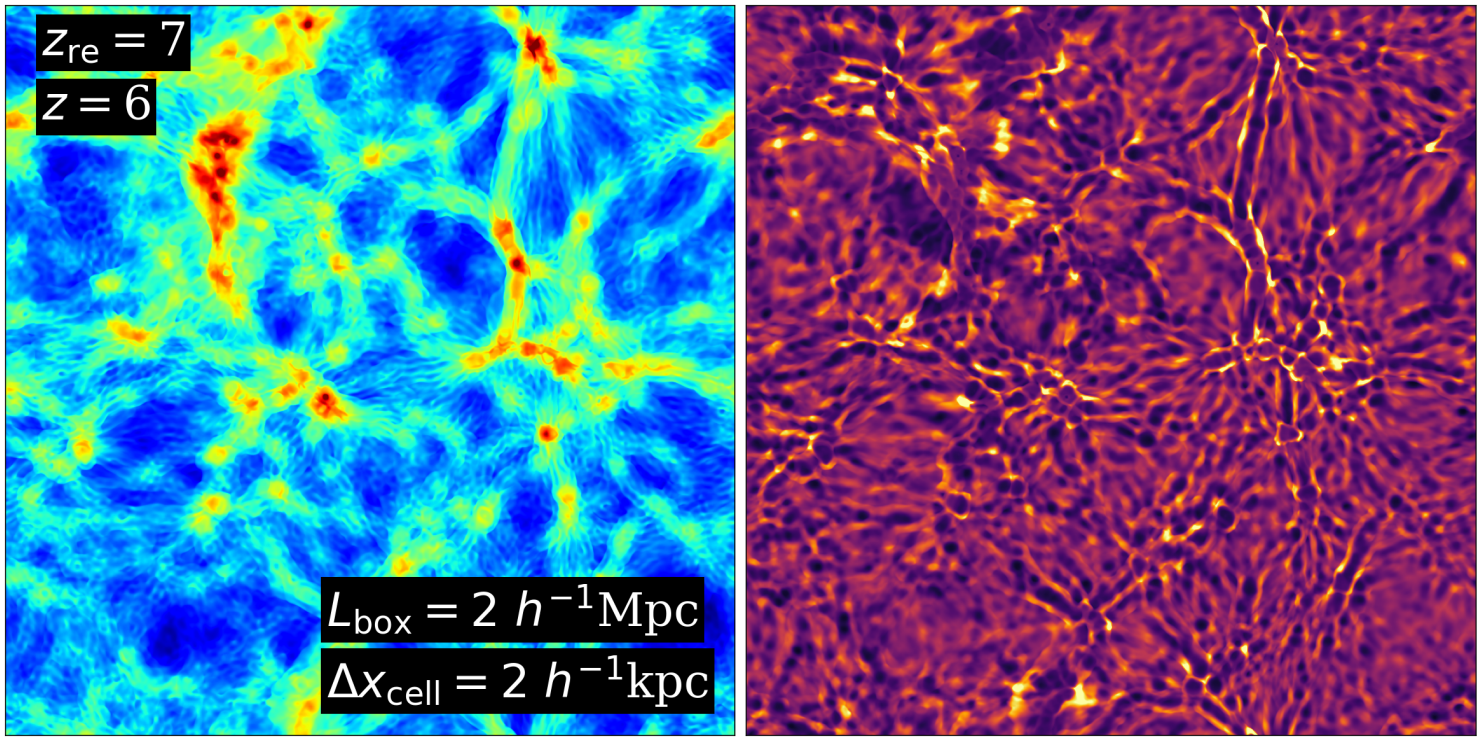
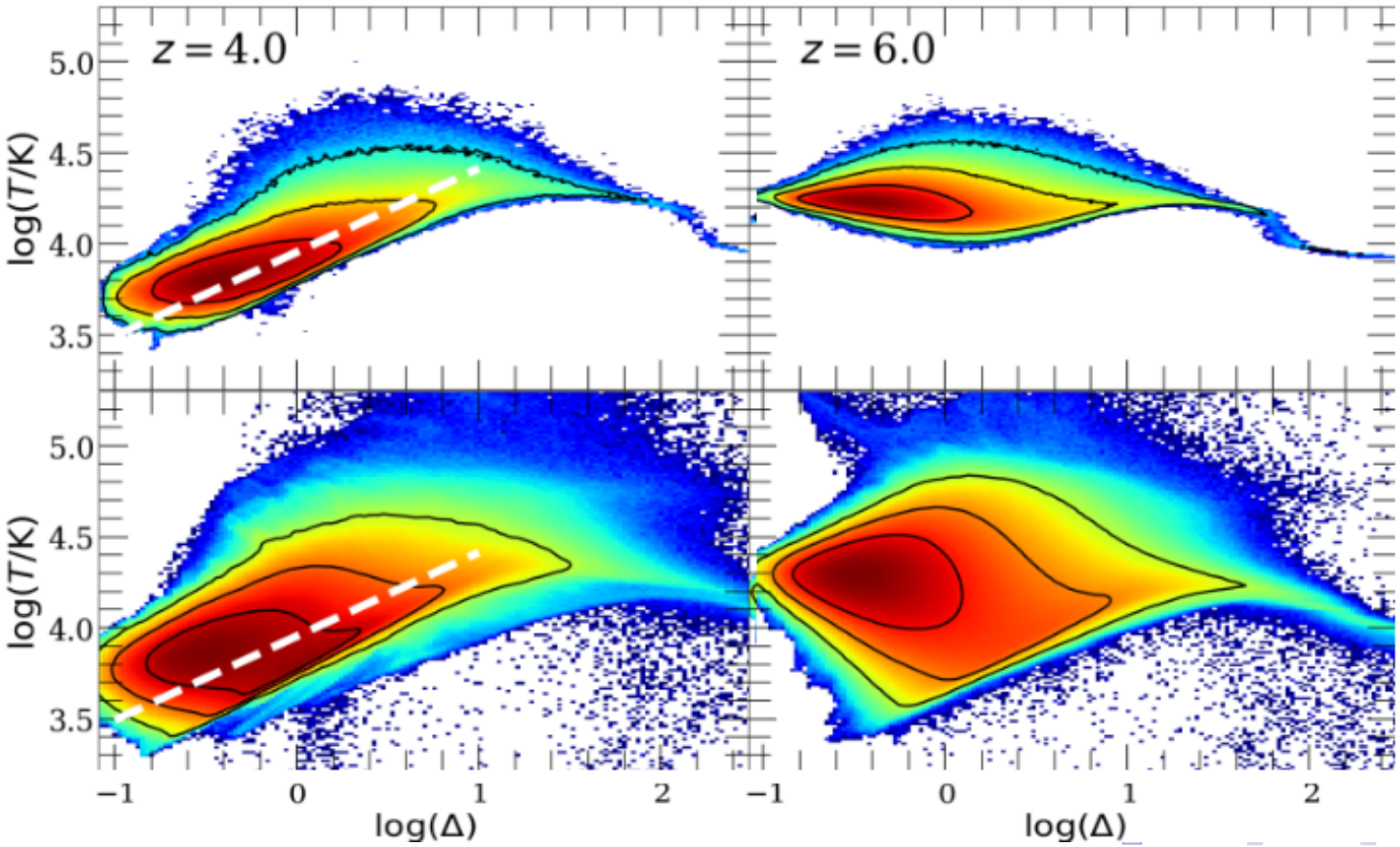


Figure: **CC**+24c

Resolution convergence of the TDR

The TDR is nowhere near converged at 8 h^{-1}ckpc resolution, even at $z = 4$ (assuming $z_{\rm reion} = 7$)



Implications

- Resolutions $\lesssim 2\ h^{-1}$ckpc may be needed to fully capture IGM thermal structure (in 40+ h^{-1}Mpc boxes for forest studies!)
- Small-scale structure depends on local reionization history $\Rightarrow$ connection to larger scales?
- Lack of convergence may affect interpretation of forest measurements (e.g. dark matter constraints)

To even smaller scales: IGM turbulence

Reasons to expect the $\sim$ *sub-kpc scale* IGM to be turbulent:

1. Intergalactic hydrogen has near-zero viscosity
2. Impulsive heating by reionization produces large pressure imbalances $\rightarrow$ driving force
3. Abundant small-scale structure throughout the IGM

Implications: IGM magnetic fields & Blazar constraints

Dynamics at sub-kpc scales

We do indeed find IGM turbulence at very small scales!

https://www.youtube.com/watch?v=gmGBoOuBhA4

Magnetic field growth via the Turbulent Dynamo

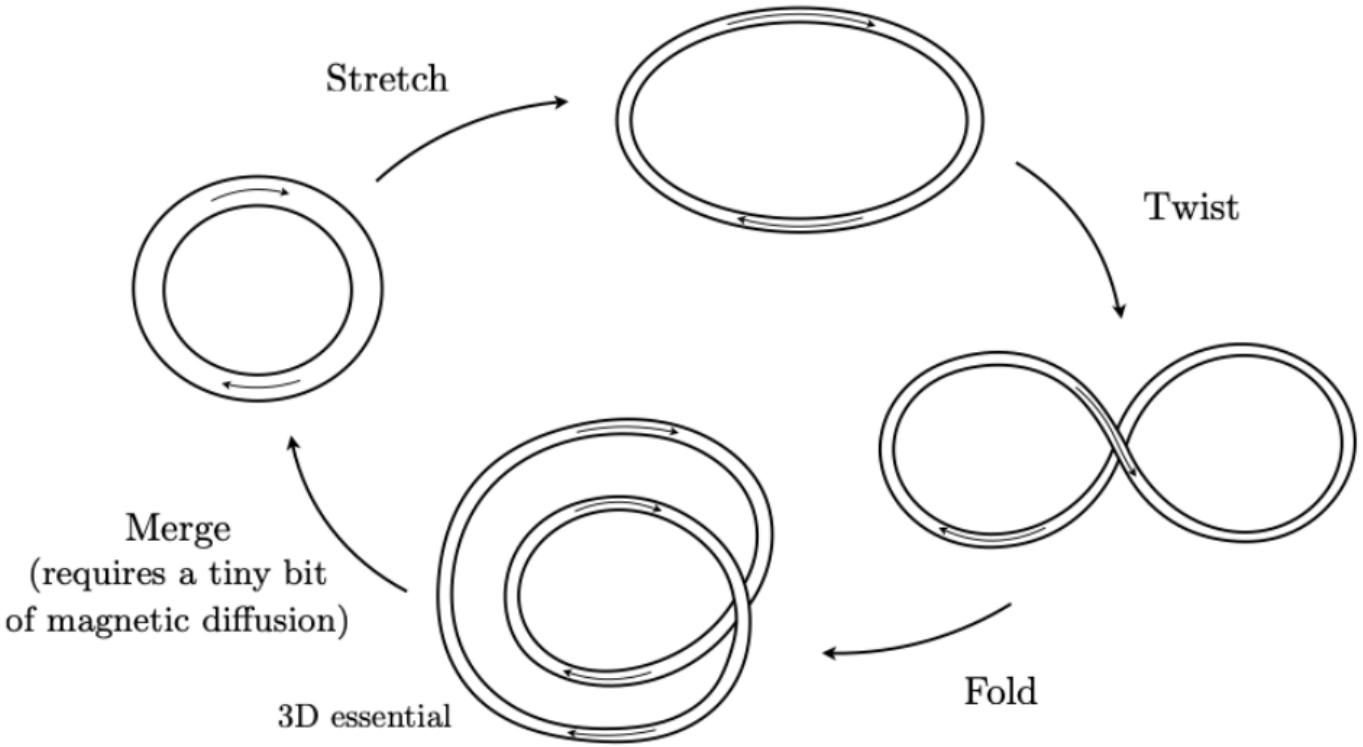
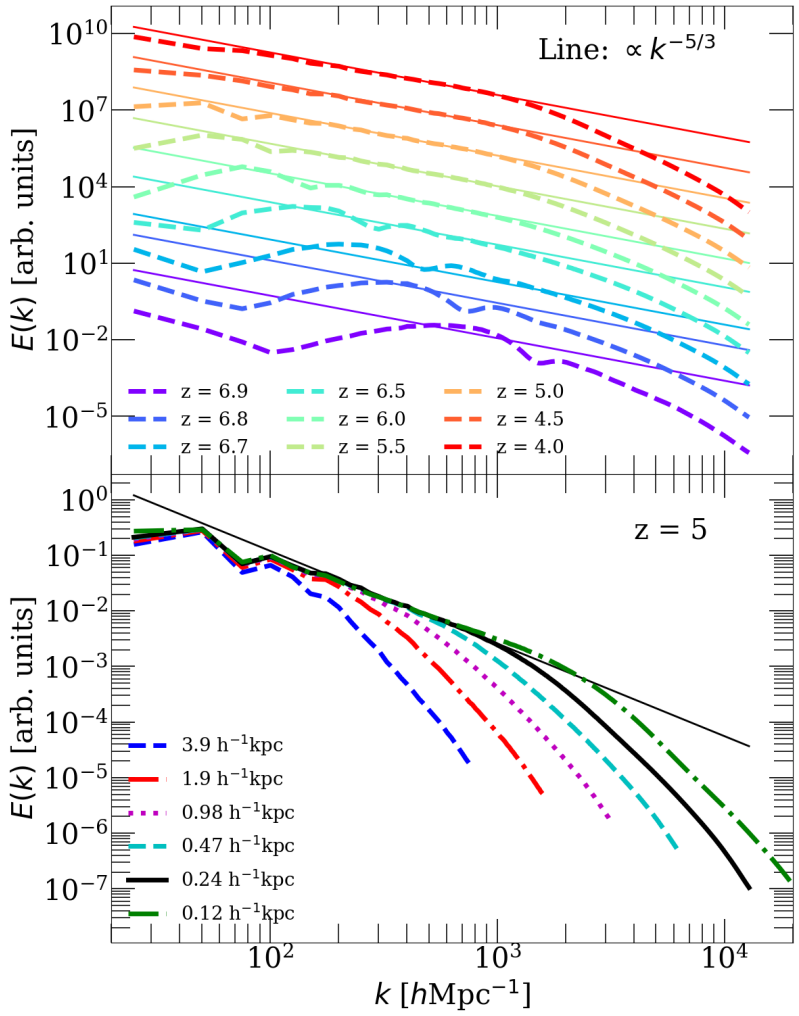


FIGURE 9. The famous stretch-twist-fold dynamo cartoon, adapted from Vainshtein & Zel'dovich (1972) and many others. (from Rincon 2019)

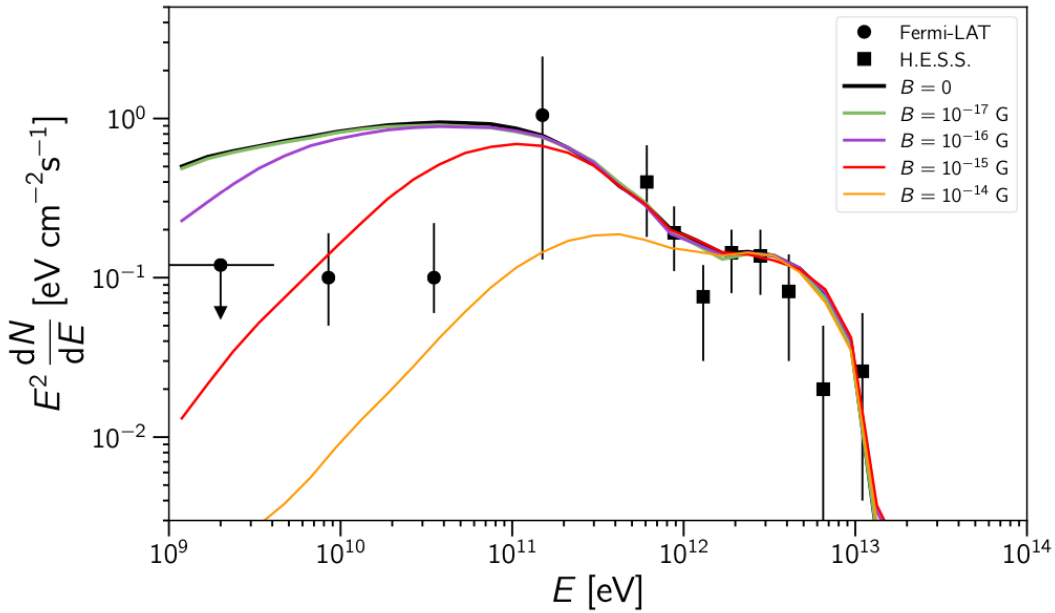
Do we see a Kolmogorov energy spectrum?

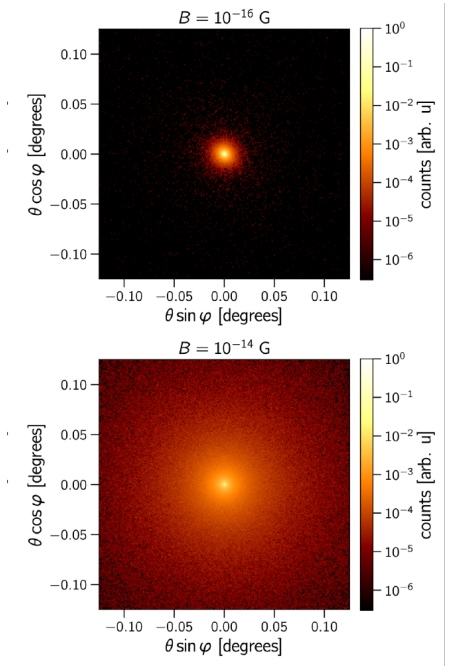
- Yes!
- Characteristic $\propto k^{-5/3}$ scaling emerges $\Delta t \approx 100$ Myr after ionization
- $300 < k/[h\mathrm{Mpc}^{-1}] < 1000$ in our highest-resolution simulation
- Diffuse hydrogen has $\mathrm{Re} \sim 10^5$, turbulence could extend to $10^{-4}\times$ driving scale (~ 1 pc!)



Could turbulence explain limits from TeV Blazars?

- Significant extended GeV emission missing from TeV blazars
- Preferred explanation is deflection by B-fields

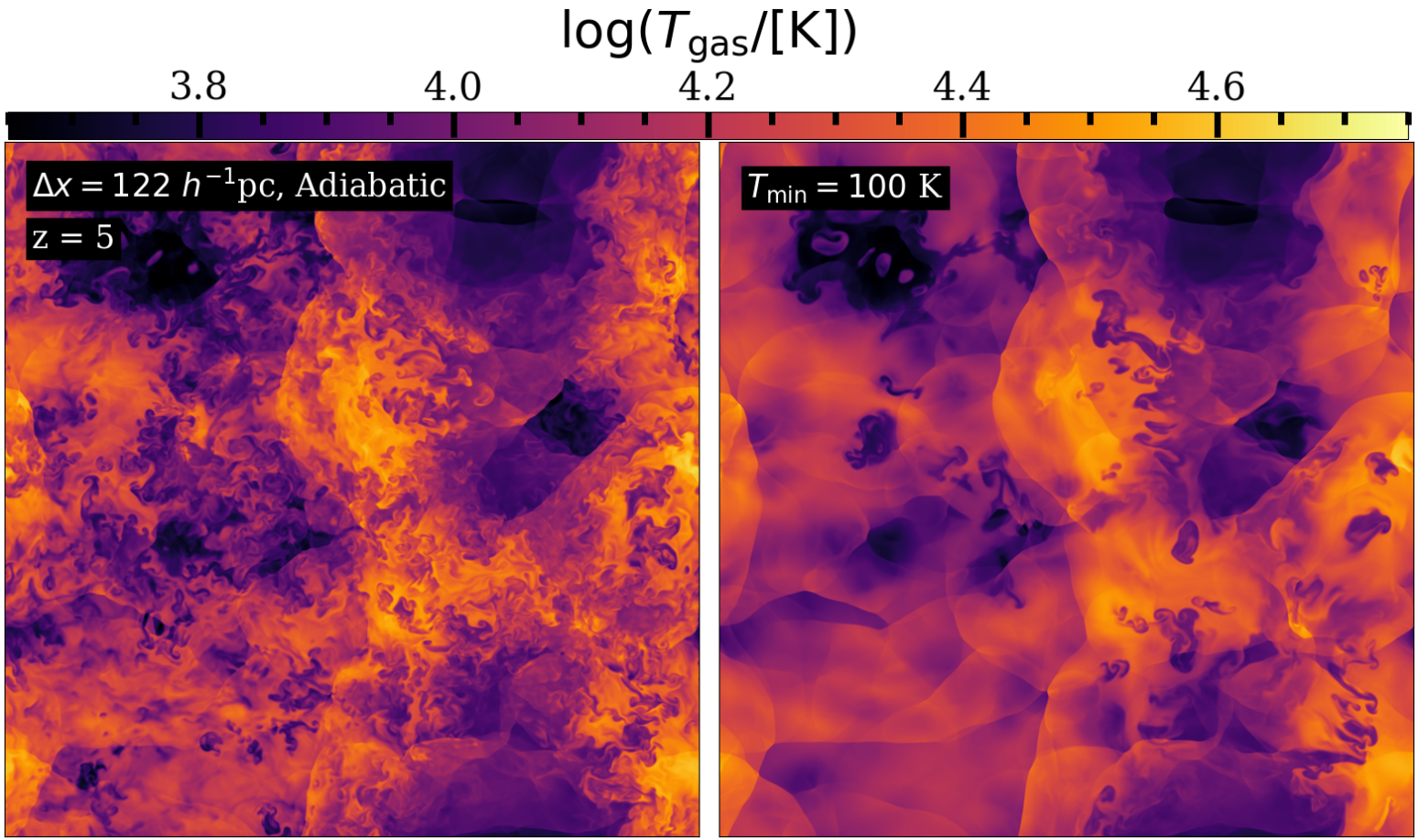




Figures: Batista & Saveliev 2021

Sensitivity to IGM pre-heating

Highly uncertain! Depends on very high-redshift X-ray sources!



Reionization & IGM Structure: Summary

- Simulating small-scale structure during reionization in a wide range of environments (SAGUARO) can help us robustly characterize IGM physics and its effects on the reionization process
- Hydrodynamic pressure smoothing driven by reionization results in a complex thermal structure on small scales, which may affect Lyα forest-based inference
- The same process may drive turbulence that could amplify intergalactic magnetic fields and explain TeV blazar observations!

Thank you!

Key papers referenced:

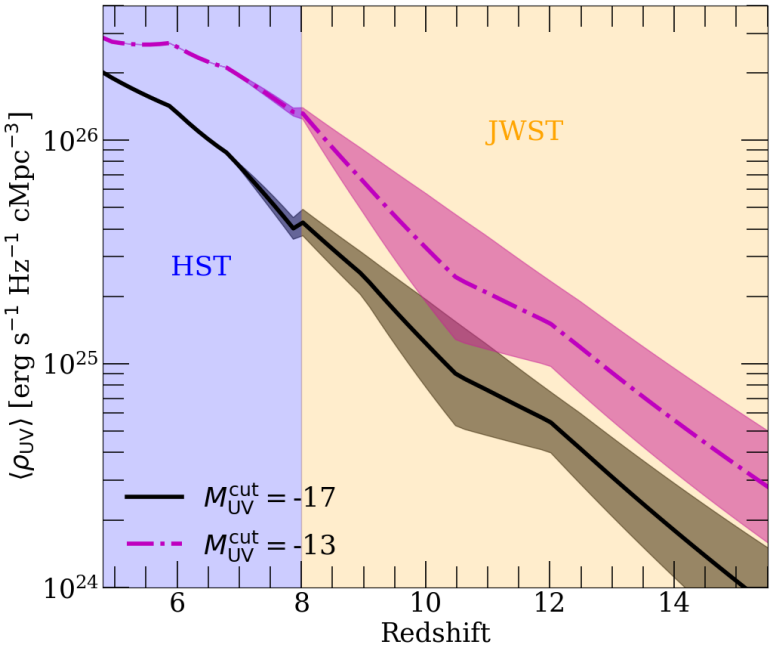
https://arxiv.org/abs/2409.04521 (FlexRT Validation)
https://arxiv.org/abs/2409.02989 (EoR History)
https://arxiv.org/abs/2311.13638 (Drop in $\dot{N}_{\rm ion}$)
https://arxiv.org/abs/2503.08778 (Measurements of $\dot{N}_{\rm ion}$)
https://arxiv.org/abs/2405.12275 (IGM Thermal Structure)

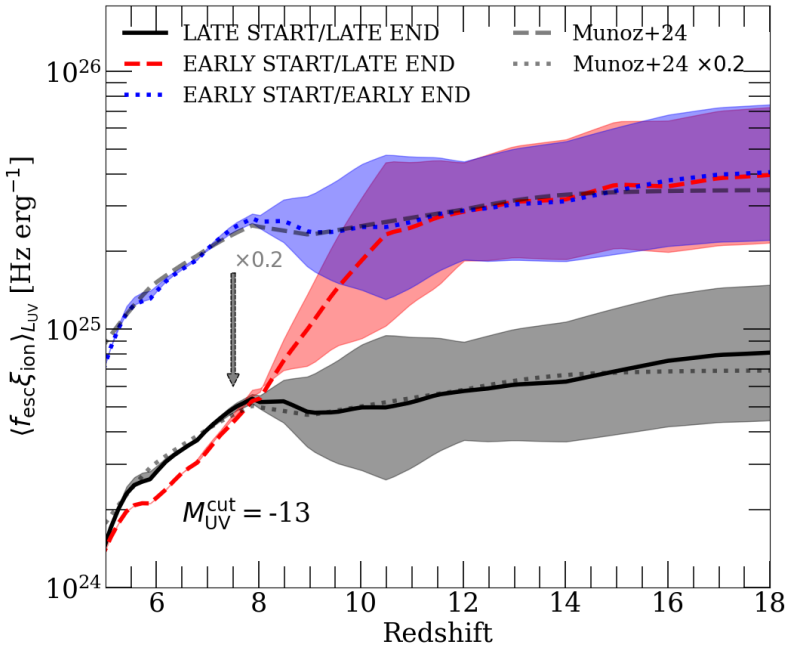
Check arXiv soon for IGM turbulence & SAGUARO papers!

Backup Slides

UVLF/Ionizing properties of galaxies

- JWST UVLF (Adams/Donnan+24) evolves rapidly at $z > 8$
- Scaled down Munoz+24 model $\rightarrow$ late start/late end
- Early start/late end $\rightarrow \sim 10\times$ evolution in $\langle f_{\rm esc}\xi_{\rm ion}\rangle$





Early start needs steeper evolution than observations suggest

Measurements of $x_{\rm HI}$ at $5 < z < 6.5$

- Constraints on the neutral fraction at $z < 6.5$ from dark gaps, dark pixels, QSO damping wings, and the forest opacity
- Recent forest damping wing constraints (Zhu+24, Spina+24) disfavor early end
- Some limits prefer an **early start**